

# SUPERSPACE

*or* One thousand and one  
lessons in supersymmetry

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## 6. QUANTUM GLOBAL SUPERFIELDS

### 6.1. Introduction to supergraphs

As we have seen in previous chapters, at the component level supersymmetric models are described by ordinary field theory Lagrangians, and their quantization and renormalization uses conventional methods. Evidently the quantum theory should be renormalized in a manner that preserves supersymmetry. Unless a manifestly supersymmetric regularization method is used, this requires applying the Ward-Takahashi identities of supersymmetry at each order of perturbation theory.

Supersymmetric models are in general less divergent than naive component power counting indicates, and this can be traced to the equality of numbers of bosonic and fermionic degrees of freedom, together with relations between coupling constants that are imposed by supersymmetry. We find that the vacuum energy (or, when (super)gravity is present, the cosmological term) receives no radiative corrections, and that, in renormalizable models, a common wave-function renormalization constant is sufficient to renormalize terms involving only scalar multiplet fields (the *no renormalization* theorem). A related result is a theorem that if the classical potential has a supersymmetric minimum (no spontaneous supersymmetry breaking), so does the effective potential to all orders of perturbation theory (no Coleman-Weinberg mechanism: see sec. 8.3.b).

Improved convergence due to supersymmetry is also evident in supergravity. For all  $N$ , the S-matrix of (extended) supergravity is finite at the first two loops; we argue in sec.7.7 that it is also finite at less than  $N - 1$  loops. In suitable supersymmetric gauges this finiteness also holds for the off-shell Green functions.

In supersymmetric theories the one-loop superconformal anomalies (trace of the energy-momentum tensor,  $\gamma$ -trace of the component supersymmetry current, and the divergence of the axial current) form a supersymmetric multiplet, the “supertrace”, so that their coefficients are equal. There exist other anomalies as well. We show in sec. 7.10 that in nonminimal  $N = 1$  supergravity ( $n \neq -\frac{1}{3}$ ), anomalies may be present in the Ward identities of local supersymmetry. Thus, in general, only minimal  $N = 1$  supergravity is consistent at the quantum level (but extended theories that have nonminimal  $N = 1$  supergravity as a submultiplet are consistent because of anomaly cancellation

mechanisms).

Superfields greatly simplify classical calculations: Supersymmetric actions can be easily constructed, and the tensor calculus of supersymmetry becomes trivial. However, the greatest advantages of superfields appear at the quantum level. There are algebraic simplifications in supersymmetric Feynman graph (“supergraph”) calculations for a number of reasons: (1) compactness of notation, (2) decrease in the number of indices (e.g., the vector field  $A_{\underline{a}}$  is hidden inside the scalar superfield  $V$ ), and (3) automatic cancellation of component graphs related by supersymmetry (which would require separate calculation in component formulations). Furthermore, the use of superfields leads to power-counting rules which explain many component results and can be used to derive additional finiteness predictions, especially when combined with supersymmetric background-field methods.

Renormalization is much simpler in the superfield formalism. Supersymmetry is manifest and, as we discuss later, *any* regularization method that preserves translational invariance in superspace will maintain it. For gauge theories we can use supersymmetric gauge-fixing terms. By contrast component Wess-Zumino gauge calculations explicitly break supersymmetry and have the disadvantage that the Ward-Takahashi identities for global supersymmetry cannot be directly applied due to their nonlinearity.

In this chapter and the next one we discuss the quantization of  $N = 1$  superfield theories. We consider classical superfield actions  $S(\Psi)$  and use functional methods to construct the generating functional  $Z(J)$  and the effective action  $\Gamma(\Psi)$ . If  $\Psi$  is a gauge field we quantize covariantly, introducing gauge-fixing terms, gauge averaging, and superfield Faddeev-Popov ghosts. We then derive Feynman rules for supergraphs using superspace propagators  $\Delta(x, x', \theta, \theta')$ . The methods are completely analogous to those for component fields, but some new features are present: We must deal with constrained (chiral) superfields, and we encounter not only  $i\partial_{\underline{a}} = p_{\underline{a}}$  operators, but also spinor derivatives  $D_{\alpha}$  acting on the arguments of propagators or external lines. We show how these operators are manipulated and how, for any graph, the  $\theta$ -integrals at each vertex can be done, leaving us with one overall  $\theta$ -integral for the whole graph (the effective action is local in  $\theta$ ), and ordinary loop-momentum integrals. At all steps of the calculations manifest supersymmetry is maintained.

We discuss next the background field method for supersymmetric Yang-Mills theories. This is similar to that for component theories, with one significant difference: The quantum-background splitting is nonlinear, reflecting the nonlinearities of the gauge transformations of the superfield  $V$ . The method simplifies many calculations and can be used to study higher-loop finiteness questions.

For supergraphs the simplest regularization procedure is to use dimensional regularization of momentum integrals *after* the contribution from a graph has been reduced to a single  $\theta$  integral. The resulting effective action, which is a (local in  $\theta$ ) functional of the external superfields, is manifestly supersymmetric. However, this regularization method corresponds to (component) regularization by dimensional reduction, which is known to be inconsistent. The superfield results, although supersymmetric, may reflect this inconsistency by exhibiting ambiguities associated with the order in which some of the  $\theta$ -integrations have been carried out. We also discuss alternative regularization procedures. Besides giving power counting rules we do not discuss the details of the renormalization of superfield theories. We work with Wick-rotated time coordinates:  $d^4x \rightarrow id^4x$ , so  $e^{-iS} \rightarrow e^S$ . (The metric  $\eta_{ab}$  has signature  $(-+++)$   $\rightarrow$   $(++++)$ , so  $\square \rightarrow +\square$ , etc. Note that in our conventions,  $i$  is opposite in sign from usual conventions: Thus positive-energy states are described by  $e^{i\omega t}$  and propagators are  $(p^2 + m^2 + i\epsilon)^{-1}$ . We further warn the reader that the gauge coupling constant  $g$  is  $\sqrt{2}$  times the usual  $g$  (see page 55).)

## 6.2. Gauge fixing and ghosts

The quantization of supersymmetric gauge theories is similar to that of ordinary gauge theories. There are two related aspects of the situation: (a) The action is invariant under gauge transformations and therefore the functional integration should be restricted to the subset of gauge inequivalent fields. (b) The kinetic operator is not invertible over the space of all field configurations so that the propagator, needed for doing perturbation theory, cannot be defined unless the set of fields is restricted. In component gauge theories, imposing an algebraic restriction explicitly in the functional integral leads to an axial gauge which breaks manifest Lorentz invariance. Alternatively, we can quantize covariantly using the Faddeev-Popov procedure: We introduce gauge fixing function(s), weighted gauges and Faddeev-Popov ghosts. In supersymmetric gauge theories the analog of the axial gauge is the Wess-Zumino gauge. In this gauge, quantization breaks manifest supersymmetry. In contrast, covariant superfield quantization maintains manifest supersymmetry.

### a. Ordinary Yang-Mills theory

For orientation we briefly recall the quantization method for ordinary Yang-Mills theory. The Yang-Mills gauge action is

$$S_{YM} = \frac{1}{g^2} \text{tr} \int d^4x \left[ -\frac{1}{8} f^{ab} f_{ab} \right] \quad , \quad f_{ab} = \partial_{[a} A_{b]} - i[A_a, A_b] \quad , \quad (6.2.1)$$

with gauge invariance under the transformation

$$A'_a \equiv A_a^\omega = e^{i\omega} [A_a + i\partial_a] e^{-i\omega} \quad , \quad (6.2.2a)$$

or, infinitesimally,

$$\delta A_a = \nabla_a \omega = \partial_a \omega + i[\omega, A_a] \quad . \quad (6.2.2b)$$

Here  $\omega$  is an element of the gauge algebra. Both  $A$  and  $\omega$  are matrices in the adjoint representation. We observe that the kinetic (quadratic) part of the Lagrangian can be written (after rescaling  $A \rightarrow gA$ ) in the form  $\frac{1}{2} A \square \Pi^T A$  where  $(\Pi^T)_{ab} = \eta_{ab} - \frac{1}{2} \partial_a \partial_b \square^{-1}$  is a transverse projection operator (see (3.11.2)).

We start with the normalized functional integral

$$Z = N' \int \mathcal{D}A_{\underline{a}} e^{S_{inv}} , \quad (6.2.3)$$

where we have included in  $S_{inv}$  possible terms with sources coupled to gauge invariant operators. We define the gauge invariant integral over the group manifold

$$\Delta_F(A_{\underline{a}}) = \int \mathcal{D}\omega \delta[F(A_{\underline{a}}^{\omega}) - f(x)] , \quad (6.2.4)$$

where  $f(x)$  is an arbitrary field-independent function, and  $F$  is a gauge-variant function such that  $F = f$  for some value of  $\omega$ . It is important to verify that this is the case. We introduce a factor of 1 in the functional integral, in the form  $\Delta_F^{-1} \Delta_F$ :

$$\begin{aligned} Z &= N' \int \mathcal{D}A_{\underline{a}} \Delta_F^{-1}(A_{\underline{a}}) \int \mathcal{D}\omega \delta[F(A_{\underline{a}}^{\omega}) - f] e^{S_{inv}} \\ &= N' \int \mathcal{D}A_{\underline{a}} \Delta_F^{-1}(A_{\underline{a}}) \int \mathcal{D}\omega \delta[F(A_{\underline{a}}) - f] e^{S_{inv}} , \end{aligned} \quad (6.2.5)$$

where the last form follows from a change of variables that is a gauge transformation, and the gauge invariance of  $\Delta_F$  and  $S$ . The  $\omega$  integral now gives a constant infinite factor that we absorb into the normalization  $N'$ , leading to the form

$$Z = N' \int \mathcal{D}A_{\underline{a}} \Delta_F^{-1}(A_{\underline{a}}) \delta[F(A_{\underline{a}}) - f] e^{S_{inv}} . \quad (6.2.6)$$

By construction  $Z$  is independent of  $F$  and  $f$ , and hence we can average over  $f$  with an arbitrary (normalized) weighting factor. In particular, if we introduce a factor  $1 = N'' \int \mathcal{D}f \exp(-\frac{1}{g^2\alpha} \text{tr} \int d^4x f^2)$ , the  $\delta(F(A) - f)$  factor can be used to carry out the integration and leads to the form

$$\begin{aligned} Z &= N' \int \mathcal{D}A_{\underline{a}} \Delta_F^{-1} e^{S_{inv} + S_{GF}} , \\ S_{GF} &= -\frac{1}{g^2\alpha} \text{tr} \int d^4x [F(A_{\underline{a}})]^2 . \end{aligned} \quad (6.2.7)$$

where we have absorbed  $N''$  into  $N'$ .

We can parametrize the gauge group by a gauge parameter  $\omega(x)$  such that  $F(A_a^{\omega}) = f(x)$  for  $\omega = 0$ . Then

$$\begin{aligned}
\Delta_F(A_{\underline{a}}) &= \int \mathcal{D}\omega \, \delta[F(A_{\underline{a}}^\omega) - f] = \int \mathcal{D}\omega \, \left[\frac{\delta F}{\delta \omega}\right]^{-1} \delta(\omega) \\
&= \int \mathcal{D}\omega \, \delta\left[\frac{\delta F}{\delta \omega} \omega\right] = \int \mathcal{D}\omega \, \mathcal{D}\omega' \, e^{\int \omega' \frac{\delta F}{\delta \omega} \omega \, dx} \, , \tag{6.2.8}
\end{aligned}$$

where we have written an integral representation for the functional  $\delta$ -function. In the second line of (6.2.8), and in the equations below,  $\frac{\delta F}{\delta \omega}$  is evaluated at  $\omega = 0$ . To obtain  $\Delta_F^{-1}$  we replace  $\omega$  and  $\omega'$  by real anticommuting (Faddeev-Popov ghost) fields  $c(x)$  and  $c'(x)$  (see sec. 3.7). Finally, we can choose for the gauge fixing function the form  $F(A_a) = \frac{1}{2} \partial^a A_{\underline{a}}$ . Then  $\frac{\delta F}{\delta \omega} \omega = \frac{1}{2} \partial^a \nabla_{\underline{a}} \omega$ , and we have

$$\begin{aligned}
Z &= N' \int \mathcal{D}A_{\underline{a}} \mathcal{D}c \mathcal{D}c' \, e^{S_{eff}} \, , \\
S_{eff} &= \frac{1}{g^2} \text{tr} \int d^4x \, [L_{inv}(A_{\underline{a}}) - \frac{1}{\alpha} F(A)^2 + ic' \frac{\delta F}{\delta \omega} c] \\
&= \frac{1}{g^2} \text{tr} \int d^4x \, [L_{inv}(A_{\underline{a}}) - \frac{1}{4\alpha} (\partial^a A_{\underline{a}})^2 + ic' \frac{1}{2} \partial^a \nabla_{\underline{a}} c] \, . \tag{6.2.9}
\end{aligned}$$

(The  $i$  is for hermiticity.) The gauge-fixing term can be written in the form  $\frac{1}{2\alpha} A \square \Pi^L A$  where  $\Pi^L = 1 - \Pi^T$  is the longitudinal projection operator  $(\Pi^L)_{\underline{a}\underline{b}} = \frac{1}{2} \partial_{\underline{a}} \partial_{\underline{b}} \square^{-1}$ . The total kinetic operator becomes  $\square(1 + (\frac{1}{\alpha} - 1)\Pi^L)$ , which is invertible: Minus its inverse (the propagator) is  $-\square^{-1}(1 + (\alpha - 1)\Pi^L)$ . In the Fermi-Feynman gauge,  $\alpha = 1$ , the propagator is  $-\square^{-1}$ .

The gauge-fixed Lagrangian, including ghosts, is invariant under the *global* BRST transformations:

$$\begin{aligned}
\delta A_{\underline{a}} &= i\xi \nabla_{\underline{a}} c \, , \\
\delta c' &= \frac{2}{\alpha} \xi F = \frac{1}{\alpha} \xi \partial_{\underline{a}} A^{\underline{a}} \, , \\
\delta c &= -\xi c^2 \, , \tag{6.2.10}
\end{aligned}$$

with constant Grassmann parameter  $\xi$ . These transformations are nilpotent:  $\delta^2$  on any

field vanishes (when the antighost equations of motion are imposed). The Ward identities for this global invariance are the Slavnov-Taylor identities of the gauge theory.

### b. Supersymmetric Yang-Mills theory

For supersymmetric Yang-Mills theory we quantize following the same procedure. We start with the functional integral for a gauge real scalar superfield  $V = V^A T_A$ , where  $T_A$  are the generators of the gauge group:

$$Z = \int \mathcal{D}V e^{S_{inv}(V)} . \quad (6.2.11)$$

Note that in supersymmetric theories the normalization factor of (6.2.3)  $N' = 1$  (see sec. 3.8.b). The action is

$$\begin{aligned} S_{inv} &= \frac{1}{g^2} \text{tr} \int d^4x d^2\theta W^2 \\ &= -\frac{1}{2g^2} \text{tr} \int d^4x d^4\theta (e^{-V} D^\alpha e^V) \bar{D}^2 (e^{-V} D_\alpha e^V) \\ &= \frac{1}{2g^2} \text{tr} \int d^4x d^4\theta [V D^\alpha \bar{D}^2 D_\alpha V + \text{higher-order terms}] . \end{aligned} \quad (6.2.12)$$

It is invariant under the gauge transformations

$$e^{V'} = e^{i\bar{\Lambda}} e^V e^{-i\Lambda} , \quad (6.2.13a)$$

or, for infinitesimal  $\Lambda$  (see (4.2.28)),

$$\delta V = L_{\frac{1}{2}V} [-i(\bar{\Lambda} + \Lambda) + \coth L_{\frac{1}{2}V} i(\bar{\Lambda} - \Lambda)] , \quad L_X Y = [X, Y] . \quad (6.2.13b)$$

In the abelian case, this is  $\delta V = i(\bar{\Lambda} - \Lambda)$ . The kinetic operator is  $\square \Pi_{\frac{1}{2}}$  with the superspin  $\frac{1}{2}$  projection operator  $\Pi_{\frac{1}{2}} = -\square^{-1} D^\alpha \bar{D}^2 D_\alpha$ , and is not invertible because it annihilates the chiral and antichiral superspin zero parts of  $V$ :  $V_0 = \Pi_0 V = \square^{-1} (D^2 \bar{D}^2 + \bar{D}^2 D^2) V$ .

We must now choose gauge-fixing functions. Corresponding to the chiral gauge parameter  $\Lambda$  we need a gauge-variant function that can be made to vanish by a suitable gauge transformation. Therefore it must have the same spin and superspin as the gauge

parameter and hence should be chosen a chiral scalar. The gauge-variant quantity  $F = \bar{D}^2 V$  is a suitable gauge-fixing function. For any chiral function  $f(x, \theta)$ , we verify that gauge transformations can be found to make  $F = f$ . For example, in the abelian case, under a gauge transformation  $F(V) \rightarrow F(V^\Lambda) = \bar{D}^2 V + i \bar{D}^2 \bar{\Lambda}$ ; if we choose  $i \bar{\Lambda} = \square^{-1} D^2 (f - \bar{D}^2 V)$ , we find  $F^\Lambda = f$ .

We define the functional determinant

$$\Delta(V) = \int \mathcal{D}\Lambda \mathcal{D}\bar{\Lambda} \delta[F(V, \Lambda, \bar{\Lambda}) - f] \delta[\bar{F}(V, \Lambda, \bar{\Lambda}) - \bar{f}] \quad . \quad (6.2.14)$$

We first write (cf. (6.2.5))

$$Z = \int \mathcal{D}V \Delta^{-1}(V) \delta[\bar{D}^2 V - f] \delta[D^2 V - \bar{f}] e^{S_{inv}} \quad . \quad (6.2.15)$$

As in (6.2.7), we average over  $f$  and  $\bar{f}$  with a weighting factor  $\int \mathcal{D}f \mathcal{D}\bar{f} \exp(-\frac{1}{g^2 \alpha} \text{tr} \int d^4 x d^4 \theta \bar{f} f)$ , and obtain the form

$$Z = \int \mathcal{D}V \Delta^{-1}(V) e^{S_{inv} + S_{GF}} \quad , \quad (6.2.16)$$

where

$$S_{GF} = -\frac{1}{\alpha g^2} \text{tr} \int d^4 x d^4 \theta (D^2 V) (\bar{D}^2 V) \quad . \quad (6.2.17)$$

We write

$$\Delta(V) = \int \mathcal{D}\Lambda \mathcal{D}\bar{\Lambda} \mathcal{D}\Lambda' \mathcal{D}\bar{\Lambda}' e^{\int d^4 x d^2 \theta \Lambda' \left( \frac{\delta F}{\delta \Lambda} \Lambda + \frac{\delta F}{\delta \bar{\Lambda}} \bar{\Lambda} \right) + \int d^4 x d^2 \bar{\theta} \bar{\Lambda}' \left( \frac{\delta \bar{F}}{\delta \Lambda} \Lambda + \frac{\delta \bar{F}}{\delta \bar{\Lambda}} \bar{\Lambda} \right)} \quad , \quad (6.2.18)$$

where we have replaced the  $\delta$ -functions involving (anti)chiral quantities by integral representations involving (anti)chiral parameters and integration measures. The variational derivatives of  $F$ ,  $\bar{F}$ , are evaluated at  $\Lambda = \bar{\Lambda} = 0$ . In the functional integral, where  $\Delta^{-1}(V)$  appears, we replace the parameters  $\Lambda, \Lambda'$  by anticommuting chiral ghost fields  $c, c'$ . Finally, we find

$$Z = \int \mathcal{D}V \mathcal{D}c \mathcal{D}c' \mathcal{D}\bar{c} \mathcal{D}\bar{c}' e^{S_{inv} + S_{GF} + S_{FP}} \quad , \quad (6.2.19)$$

where

$$\begin{aligned}
S_{FP} &= i \operatorname{tr} \int d^4x d^2\theta c' \bar{D}^2(\delta V) + i \operatorname{tr} \int d^4x d^2\bar{\theta} \bar{c}' D^2(\delta V) \\
&= \operatorname{tr} \int d^4x d^4\theta (c' + \bar{c}') L_{\frac{1}{2}V} [(c + \bar{c}) + \coth L_{\frac{1}{2}V} (c - \bar{c})] . \tag{6.2.20}
\end{aligned}$$

Integrating by parts, we can write  $(D^2V)(\bar{D}^2V) = \frac{1}{2} V(D^2\bar{D}^2 + \bar{D}^2D^2)V = \frac{1}{2} V\Box\Pi_0V$ .

The quadratic part of the gauge field action has now the form

$$-\frac{1}{2} V\Box(\Pi_{\frac{1}{2}} + \alpha^{-1}\Pi_0)V = -\frac{1}{2} V\Box[1 + (\alpha^{-1} - 1)\Pi_0]V , \tag{6.2.21}$$

and the operator is invertible. To avoid  $\Box^{-2}$  terms in the propagator and thus bad infrared behavior, we choose the supersymmetric Fermi-Feynman gauge  $\alpha = 1$ , which leads to a simple  $\Box^{-1}$  propagator. (The  $-$  sign in (6.2.21) leads to the usual kinetic term for the component gauge field:  $-\int d^4\theta V\Box V \sim A_a\Box A^a$ .)

The quadratic part of the ghost action has the form

$$S_{FP}^{(2)} = \operatorname{tr} \int d^4x d^4\theta (c' + \bar{c}')(c - \bar{c}) = \operatorname{tr} \int d^4x d^4\theta (\bar{c}'c - c'\bar{c}) . \tag{6.2.22}$$

The chiral and antichiral  $c'c$  and  $\bar{c}'\bar{c}$  terms vanish when integrated with  $d^4\theta$  and have been dropped. (Such terms cannot be dropped in the presence of supergravity fields: see, for example, (5.5.16)).

The total action is invariant under superfield BRST transformations. These take the form

$$\begin{aligned}
\delta V &= \delta_\Lambda V|_{\Lambda=i\xi c} = \xi L_V[(c + \bar{c}) + \coth L_{\frac{1}{2}V} (c - \bar{c})] , \\
\delta c' &= \frac{1}{\alpha} \xi \bar{D}^2 \bar{F} = \frac{1}{\alpha} \xi \bar{D}^2 D^2 V , \quad \delta \bar{c}' = \frac{1}{\alpha} \xi D^2 F = \frac{1}{\alpha} \xi D^2 \bar{D}^2 V , \\
\delta c &= -\xi c^2 , \quad \delta \bar{c} = -\xi \bar{c}^2 , \tag{6.2.23}
\end{aligned}$$

and the invariance can be used to derive the Slavnov-Taylor identities of the theory.

Before performing perturbation expansions, we rescale  $V \rightarrow gV$ . Then all quadratic terms are  $O(g^0)$ , cubic terms are  $O(g)$ , etc. We rescale back  $gV \rightarrow V$  in the effective action  $\Gamma$ . Alternatively, we simply provide each graph with a factor  $(g^2)^{L-1}$ ,

where  $L$  is the number of loops.

### c. Other gauge multiplets

We give two other examples of the gauge-fixing procedure: For a chiral superfield  $\Phi$ , the solution of the chirality constraint,  $\Phi = \bar{D}^2 \Psi$ , gives the kinetic action

$$S_{inv} = \int d^4x d^4\theta \bar{\Psi} D^2 \bar{D}^2 \Psi \quad , \quad (6.2.24)$$

and introduces the gauge invariance

$$\delta \Psi = \bar{D}^{\dot{\alpha}} \bar{\omega}_{\dot{\alpha}} \quad , \quad \delta \bar{\Psi} = D^{\alpha} \omega_{\alpha} \quad , \quad (6.2.25)$$

for an arbitrary spinor parameter  $\omega_{\alpha}$  (see (4.5.1-4)). Suitable gauge fixing functions are the linear spinor superfields

$$F_{\alpha} = D_{\alpha} \Psi \quad , \quad \bar{F}_{\dot{\alpha}} = \bar{D}_{\dot{\alpha}} \bar{\Psi} \quad . \quad (6.2.26)$$

To obtain a convenient gauge fixing term we average with  $\bar{f}^{\dot{\alpha}} M_{\alpha\dot{\alpha}} f^{\alpha}$ , where

$$M_{\alpha\dot{\alpha}} = D_{\alpha} \bar{D}_{\dot{\alpha}} + \frac{3}{4} \bar{D}_{\dot{\alpha}} D_{\alpha} \quad . \quad (6.2.27)$$

This leads to

$$S_{inv} + S_{GF} = \int d^4x d^4\theta \bar{\Psi} \square \Psi \quad , \quad (6.2.28)$$

and a standard  $p^{-2}$  propagator.

A second example is for the action of the chiral spinor superfield  $\Phi_{\alpha}$  that describes the tensor multiplet (4.4.46):

$$S_{inv} = -\frac{1}{2} \int d^4x d^4\theta G^2 = -\frac{1}{8} \int d^4x d^4\theta (D^{\alpha} \Phi_{\alpha} + \bar{D}^{\dot{\alpha}} \bar{\Phi}_{\dot{\alpha}})^2 \quad , \quad (6.2.29)$$

with gauge invariance under

$$\delta \Phi_{\alpha} = i \bar{D}^2 D_{\alpha} K \quad , \quad K = \bar{K} \quad . \quad (6.2.30)$$

A suitable gauge fixing function is

$$F = -i \frac{1}{2} (D^{\alpha} \Phi_{\alpha} - \bar{D}^{\dot{\alpha}} \bar{\Phi}_{\dot{\alpha}}) \quad , \quad (6.2.31)$$

where  $F$  is linear. The gauge-fixed action

$$\begin{aligned}
 S_{inv} + S_{GF} &= \frac{1}{2} \int d^4x d^4\theta \left[ -G^2 + \frac{1}{\alpha} F^2 \right] \\
 &= -\frac{1}{4} \int d^4x d^2\theta \Phi^\alpha \square \left[ \frac{1}{2} (1 + \mathbf{K}) + \frac{1}{\alpha} \frac{1}{2} (1 - \mathbf{K}) \right] \Phi_\alpha + h.c. \\
 &= \frac{1}{2} \int d^4x d^4\theta \left[ -\frac{1}{2} (1 + \alpha) \left( \frac{1}{2} \Phi^\alpha D^2 \Phi_\alpha + h.c. \right) + \frac{1}{2} (1 - \alpha) \bar{\Phi}^{\dot{\alpha}} i \partial^\alpha_{\dot{\alpha}} \Phi_\alpha \right]
 \end{aligned} \tag{6.2.32}$$

(with  $\mathbf{K}$  as in sec 3.11) takes two convenient forms:

For  $\alpha = 1$ ,

$$S_{inv} + S_{GF} = -\frac{1}{4} \int d^4x d^4\theta (\Phi^\alpha D^2 \Phi_\alpha + \bar{\Phi}^{\dot{\alpha}} \bar{D}^2 \bar{\Phi}_{\dot{\alpha}}) ; \tag{6.2.33}$$

for  $\alpha = -1$ ,

$$S_{inv} + S_{GF} = \frac{1}{2} \int d^4x d^4\theta \bar{\Phi}^{\dot{\alpha}} i \partial^\alpha_{\dot{\alpha}} \Phi_\alpha \tag{6.2.34}$$

(cf. (3.8.36)).

### 6.3. Supergraph rules

Given an action  $S(\Psi)$ , we define the generating functional for Green functions

$$Z(J) = \int \mathcal{D}\Psi e^{S(\Psi) + \int J\Psi} , \quad (6.3.1)$$

where  $J$  is a source of the same type as the field  $\Psi$  (general if  $\Psi$  is general, chiral if  $\Psi$  is chiral, etc.). The generating functional of *connected* Green functions is

$$W(J) = \ln Z(J) . \quad (6.3.2)$$

The expectation value of the field  $\Psi$  or the “classical field”  $\hat{\Psi}$  in the presence of the source is

$$\hat{\Psi}(J) = \frac{\delta W}{\delta J} . \quad (6.3.3)$$

This relation can be inverted to give  $J(\hat{\Psi})$ . The effective action  $\Gamma(\hat{\Psi})$ , the generating functional of one particle irreducible graphs, is defined by a functional Legendre transform

$$\Gamma(\hat{\Psi}) = W[J(\hat{\Psi})] - \int J(\hat{\Psi})\hat{\Psi} . \quad (6.3.4)$$

In this section we derive the Feynman rules for the perturbative expansion of the effective action. The derivation of the Feynman rules for unconstrained superfields presents few surprises. Instead of having  $d^4x$  integrals we have  $d^4x d^4\theta$  integrals. Propagators are obtained from the inverses of the kinetic operators, and vertices can be read directly from the interaction terms. However, for chiral superfields the Feynman rules reflect the chirality constraints.

#### a. Derivation of Feynman rules

We begin by deriving the rules for the real scalar gauge superfield. The gauge fixed action (in the Fermi-Feynman gauge,  $\alpha = 1$ ) reads

$$S_V = \text{tr} \int d^4x d^4\theta \left[ -\frac{1}{2} V \square V + \frac{1}{2} [V, (D^\alpha V)] (\bar{D}^2 D_\alpha V) + \cdots \right] . \quad (6.3.5)$$

The Feynman rules can be read directly from this expression: The propagator is minus

the inverse of the kinetic operator,  $\square^{-1}\delta^4(x-x')\delta^4(\theta-\theta')$  or, in momentum space,  $-p^{-2}\delta^4(\theta-\theta')$ . Since the spinor derivatives  $D$  contain explicit  $\theta$ 's, we do *not* Fourier transform with respect to the  $\theta$  variables. (If one does Fourier transform with respect to  $\theta$ , there is little change in the Feynman rules.) Vertices can be read from the interaction terms. Thus, the cubic term  $\frac{1}{2}\text{tr}[V, (D^\alpha V)](\bar{D}^2 D_\alpha V)$  leads to a three-point vertex with factors of  $D^\alpha$  and  $\bar{D}^2 D_\alpha$  acting on two of the lines, and a group theory factor. In addition we integrate over  $x$ 's and  $\theta$ 's at each vertex or, equivalently, over loop momenta and over  $\theta$ 's at each vertex.

These rules can also be obtained by starting with the functional integral:

$$\begin{aligned} Z(J) &= \int \mathcal{D}V \, e^{\int [-\frac{1}{2}V \square V + \mathcal{L}_{int}(V) + JV]} \\ &= e^{\int \mathcal{L}_{int}\left(\frac{\delta}{\delta J}\right)} \int \mathcal{D}V \, e^{\int [-\frac{1}{2}V \square V + JV]} \\ &= e^{\int \mathcal{L}_{int}\left(\frac{\delta}{\delta J}\right)} e^{\frac{1}{2} \int J \square^{-1} J} \quad , \end{aligned} \tag{6.3.6}$$

where in the last step we have performed the Gaussian integral over  $V$ . The Feynman rules can be obtained using  $\frac{\delta J(x, \theta)}{\delta J(x', \theta')} = \delta^4(x-x')\delta^4(\theta-\theta')$  and expanding the exponentials in power series. Thus, we obtain factors of  $\mathcal{L}_{int}\left(\frac{\delta}{\delta J}\right)$  corresponding to vertices, and the  $\frac{\delta}{\delta J}$  operators, when acting on the factors of  $J \square^{-1} J$  remove the  $J$ 's and produce propagators  $\square^{-1}$  connecting the vertices. The result is exactly as for ordinary field theory, with the additional feature of  $d^4\theta$  integrals at each vertex, and additional  $\delta^4(\theta-\theta')$  factors in each propagator.

Chiral scalar superfields usually have a *kinetic* action (with chiral sources  $j, \bar{j}$ ) of the form

$$\begin{aligned} S^{(2)} &= \int d^4x \, d^4\theta \, \bar{\Phi} \Phi - \frac{1}{2} \int d^4x \, d^2\theta \, m \Phi^2 - \frac{1}{2} \int d^4x \, d^2\bar{\theta} \, m \bar{\Phi}^2 \\ &\quad + \int d^4x \, d^2\theta \, j \Phi + \int d^4x \, d^2\bar{\theta} \, \bar{j} \bar{\Phi} \quad . \end{aligned} \tag{6.3.7}$$

To perform the Gaussian integration, we rewrite chiral integrals as integrals over full

superspace. A chiral integral

$$I_c = \int d^4x d^2\theta FG \quad , \quad (6.3.8)$$

where  $F$  and  $G$  are *arbitrary* chiral expressions, can be rewritten as

$$I_c = \int d^4x d^4\theta F \square^{-1} D^2 G \quad , \quad (6.3.9)$$

using  $\square^{-1} \bar{D}^2 D^2 G = G$  (3.4.10) and  $\int d^4x d^4\theta = \int d^4x d^2\theta \bar{D}^2$ . The Gaussian integral can be rewritten as

$$e^{W_0(j)} \equiv \int D\Phi D\bar{\Phi} \exp \int d^4x d^4\theta \left[ \frac{1}{2} (\Phi \bar{\Phi}) \mathbf{O} \begin{pmatrix} \Phi \\ \bar{\Phi} \end{pmatrix} + (\Phi \bar{\Phi}) \begin{pmatrix} \square^{-1} D^2 j \\ \square^{-1} \bar{D}^2 \bar{j} \end{pmatrix} \right] \quad (6.3.10)$$

where

$$\mathbf{O} = \begin{pmatrix} -\frac{mD^2}{\square} & 1 \\ 1 & -\frac{m\bar{D}^2}{\square} \end{pmatrix} . \quad (6.3.11)$$

The inverse of  $\mathbf{O}$  is

$$\mathbf{O}^{-1} = \begin{pmatrix} \frac{m\bar{D}^2}{\square - m^2} & 1 + \frac{m^2 \bar{D}^2 D^2}{\square(\square - m^2)} \\ 1 + \frac{m^2 D^2 \bar{D}^2}{\square(\square - m^2)} & \frac{mD^2}{\square - m^2} \end{pmatrix} . \quad (6.3.12)$$

Performing the integral we obtain

$$W_0(j) = \int d^4x d^4\theta \left[ -\bar{j} \frac{1}{\square - m^2} j - \frac{1}{2} \left( j \frac{mD^2}{\square(\square - m^2)} j + h.c. \right) \right] . \quad (6.3.13)$$

For a general interaction Lagrangian  $\mathcal{L}_{int}(\Phi, \bar{\Phi})$  we can write

$$Z(j) = e^{\int d^4x d^4\theta \mathcal{L}_{int} \left( \frac{\delta}{\delta j}, \frac{\delta}{\delta \bar{j}} \right)} e^{W_0(j)} \quad , \quad (6.3.14)$$

and the Feynman rules can be obtained from this expression. Since  $\frac{\delta j(x, \theta)}{\delta j(x', \theta')} = \bar{D}^2 \delta^4(\theta - \theta') \delta^4(x - x')$  (3.8.10), there is an operator  $\bar{D}^2$  acting on each chiral

field line leaving a vertex. Similarly, there is an operator  $D^2$  acting on each antichiral line leaving a vertex. However, at a purely chiral vertex, e.g.  $\int d^2\theta \Phi^n$ , we use one of these factors to convert the  $d^2\theta$  integral to a  $d^4\theta$  integral. Therefore at such vertices we omit one factor of  $\bar{D}^2$ .

We now summarize the Feynman rules for interacting gauge and chiral superfields.

(a) Propagators:

$$VV: \quad -\frac{1}{p^2} \delta^4(\theta - \theta') \quad , \quad (6.3.15a)$$

$$\bar{\Phi}\Phi: \quad \frac{1}{p^2 + m^2} \delta^4(\theta - \theta') \quad , \quad (6.3.15b)$$

$$\Phi\Phi: \quad -\frac{mD^2}{p^2(p^2 + m^2)} \delta^4(\theta - \theta') \quad , \quad (6.3.15c)$$

$$\bar{\Phi}\bar{\Phi}: \quad -\frac{m\bar{D}^2}{p^2(p^2 + m^2)} \delta^4(\theta - \theta') \quad . \quad (6.3.15d)$$

In the massive case, the  $p^{-2}$  factors in the  $\Phi\Phi$  and  $\bar{\Phi}\bar{\Phi}$  propagators are always canceled by numerator factors (e.g., for  $\Phi\Phi$  the vertices give  $\bar{D}^2$  factors and, as we discuss later, we obtain  $\bar{D}^2 D^2 \bar{D}^2 = -p^2 \bar{D}^2$ ). In the massless case these propagators are absent.

(b) Vertices: These are read directly from the interaction Lagrangian, with the additional feature that for each chiral or antichiral line leaving a vertex there is a factor  $\bar{D}^2$  or  $D^2$  acting on the corresponding propagator, and the rule that at purely chiral or antichiral vertices we omit one  $\bar{D}^2$  or  $D^2$  factor from among the ones acting on the propagators.

(c) We integrate over  $d^4\theta$  at each vertex, and in momentum space we have loop-momentum integrals  $\int d^4p (2\pi)^{-4}$  for each loop, and an overall factor  $(2\pi)^4 \delta(\sum k_{ext})$ .

(d) To obtain the effective action  $\Gamma$ , we compute one-particle-irreducible graphs. For each external line with outgoing momentum  $k_i$ , we multiply by a factor  $\int d^4k_i (2\pi)^{-4} \Psi(k_i)$  where  $\Psi$  stands for any of the fields in the effective action. For each

external chiral or antichiral line, we have a  $\Phi$  or  $\bar{\Phi}$  factor, but no  $\bar{D}^2$  or  $D^2$  factors.

(e) Finally, there may be symmetry factors associated with certain graphs.

An alternative derivation of the Feynman rules for chiral superfields can be obtained by solving the chirality constraints in terms of an unconstrained field (see sec. 4.5a):

$$\Phi = \bar{D}^2 \Psi \quad , \quad \bar{\Phi} = D^2 \bar{\Psi} \quad , \quad (6.3.16)$$

where  $\Psi$  is a general, complex scalar superfield. The action, including source terms, becomes

$$\begin{aligned} S = & \int d^4x d^4\theta [(D^2 \bar{\Psi})(\bar{D}^2 \Psi) + \mathcal{L}_{int}(\bar{D}^2 \Psi, D^2 \bar{\Psi})] \\ & + \int d^4x d^2\theta (\bar{D}^2 \Psi)(-\frac{1}{2} m \bar{D}^2 \Psi + j) + h.c. \quad . \end{aligned} \quad (6.3.17)$$

Chiral integrals can be rewritten as full integrals by using up a  $\bar{D}^2$  factor. We recall that in terms of  $\Psi$  we have an abelian gauge invariance,  $\Psi \rightarrow \Psi + \bar{D}^{\dot{\alpha}} \bar{\omega}_{\dot{\alpha}}$  (4.5.4). Consequently, the kinetic operator appearing in the action,  $\bar{\Psi} D^2 \bar{D}^2 \Psi$ , is not invertible. As discussed in sec. 6.2, we can fix the gauge and arrive at an invertible quadratic action (the ghosts decouple)

$$S^{(2)} = \int d^4x d^4\theta \frac{1}{2} (\Psi \quad \bar{\Psi}) \begin{pmatrix} -m D^2 & \square \\ \square & -m \bar{D}^2 \end{pmatrix} \begin{pmatrix} \Psi \\ \bar{\Psi} \end{pmatrix} \quad . \quad (6.3.18)$$

The Feynman rules are now the naive ones and are identical to the ones we have obtained before (after using the  $D^2$ ,  $\bar{D}^2$  factors at the vertices to simplify the propagators, obtained from (6.3.12)). In particular, from  $\mathcal{L}_{int}(\bar{D}^2 \Psi, D^2 \bar{\Psi})$ , we again find factors of  $\bar{D}^2$ ,  $D^2$  acting on the propagators, except that one such factor is missing at purely (anti)chiral vertices, since we convert everywhere to full  $d^4\theta$  integrals.

It is simple to obtain the supergraph rules for the tensor multiplet, with gauge-invariant action

$$S = \int d^4x d^4\theta f(G) \quad , \quad f(G) = -\frac{1}{2} G^2 + \dots, \quad (6.3.19)$$

with  $\Phi\Phi$  propagator  $-2p^{-4} \delta_{\alpha}^{\beta} D^2 \delta^4(\theta - \theta')$  (and the hermitian conjugate for  $\bar{\Phi}\bar{\Phi}$ ) from

(6.2.33). However, there is a much simpler form of the rules which resembles the rules for the scalar multiplet (to which the tensor multiplet is on-shell equivalent by a duality transformation: see sec. 4.4.c.2). We first note that the vertex at either end of a  $\Phi\Phi$  propagator has a  $D$  at the vertex (from  $f(G)$  with  $G = D\Phi + \bar{D}\bar{\Phi}$ ), and next to it a  $\bar{D}^2$  (as occurs at the end of any chiral propagator, rule (b) above; this kills the  $\bar{D}\bar{\Phi}$  part of the vertex). Also note that the spinor index at the vertex contracts directly with the corresponding spinor index of the propagator (because the vertex is a function of only  $G = \frac{1}{2}D_\alpha\Phi^\alpha + h.c.$ ). Contracting these spinor indices, and integrating by parts all  $D$ 's from the vertices onto the propagators, we obtain the same expression for the  $\Phi\Phi$  and  $\bar{\Phi}\bar{\Phi}$  propagators (with the same vertices), which can now be added together (i.e., the total contribution from graphs with both types of propagators is the same as that from only one type, but with an overall factor of 2 for each propagator). The rules are thus cast into the following form: All vertices are now simply *constants*, read from the expansion of  $f(G)$  in  $G$ . There is only one type of propagator, with no spinor indices, which is

$$-\frac{1}{p^2}D^\alpha\bar{D}^2D_\alpha\delta^4(\theta-\theta') = -\Pi_{\frac{1}{2}}\delta^4(\theta-\theta') \quad . \quad (6.3.20)$$

(The algebra from the various contributing factors is  $D\bar{D}^2D^2\bar{D}^2D = D\bar{D}^2D\Box$ .) Each external line gets a factor of  $G$ . If we were to perform the same rearrangement of vertex factors for the supergraphs of the dual scalar-multiplet theory, we would obtain  $\Pi_0$  instead of  $\Pi_{\frac{1}{2}}$ , the external line factors would be  $\Phi + \bar{\Phi}$ , and the constants at the vertices would be obtained from  $\tilde{f}(\Phi + \bar{\Phi})$  in terms of the function  $\tilde{f}$  dual to  $f$  (see again sec. 4.4.c.2). The on-shell equivalence then follows from the fact that the combinatorics resulting from using  $\Pi_{\frac{1}{2}} = 1 - \Pi_0$ , with a propagator 1 collapsing to a point (in  $\theta$  and  $x$ ), performs the duality, where for the external lines  $G = \Phi + \bar{\Phi}$  on shell.

### b. A sample calculation

We now give an example in a theory of a massless chiral superfield  $\Phi$  interacting with a gauge superfield  $V$ . We compute the one-loop contribution from the chiral superfield to the  $V$  two-point function. The relevant interaction is obtained from  $\bar{\Phi}e^V\Phi = \bar{\Phi}\Phi + \bar{\Phi}V\Phi + \dots$ . We find a contribution to the effective action, according to our rules and Fig. 6.3.1,

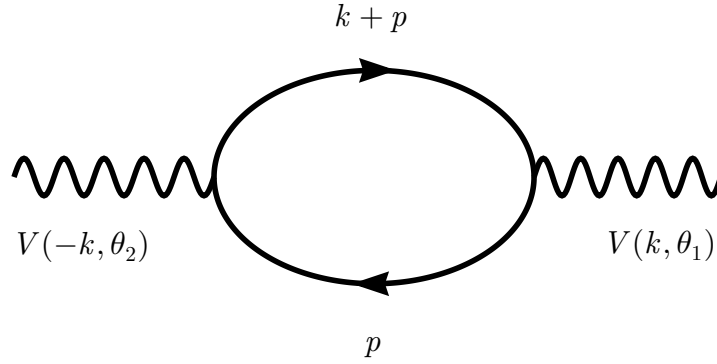


Fig. 6.3.1

$$\frac{1}{2} \int \frac{d^4 k}{(2\pi)^4} d^4 \theta_1 d^4 \theta_2 V(-k, \theta_2) V(+k, \theta_1) \cdot \int \frac{d^4 p}{(2\pi)^4} \frac{D_1^2 \delta^4(\theta_1 - \theta_2) \overleftarrow{\bar{D}}_2^2}{p^2} \frac{D_2^2 \delta^4(\theta_2 - \theta_1) \overleftarrow{\bar{D}}_1^2}{(p+k)^2} . \quad (6.3.21)$$

Note that in the above expression

$$D_{1\alpha} = \frac{\partial}{\partial \theta_1^\alpha} + \frac{1}{2} \bar{\theta}_1^{\dot{\alpha}} p_{\alpha\dot{\alpha}} , \quad \bar{D}_{1\dot{\alpha}} = \frac{\partial}{\partial \bar{\theta}_1^{\dot{\alpha}}} - \frac{1}{2} \theta_1^\alpha (k+p)_{\alpha\dot{\alpha}} ,$$

$$D_{2\alpha} = \frac{\partial}{\partial \theta_2^\alpha} + \frac{1}{2} \bar{\theta}_2^{\dot{\alpha}} (k+p)_{\alpha\dot{\alpha}} , \quad \bar{D}_{2\dot{\alpha}} = \frac{\partial}{\partial \bar{\theta}_2^{\dot{\alpha}}} - \frac{1}{2} \theta_2^\alpha p_{\alpha\dot{\alpha}} . \quad (6.3.22)$$

Although we do not indicate the momentum dependence explicitly, it is implicit that the momentum is that *leaving* the vertex through the propagator on which the operators act. (From the  $\bar{\Phi} V^2 \Phi$  interaction term we also obtain a tadpole-type diagram; its contribution cancels a similar contribution from the diagram we are considering, or vanishes if we use dimensional regularization).

The  $D$ 's can be manipulated like ordinary derivatives. They obey a Leibnitz rule, and a “transfer” rule

$$\delta^4(\theta_1 - \theta_2) \overleftarrow{\bar{D}}_{\dot{\alpha}2}(p) = - \overleftarrow{\bar{D}}_{\dot{\alpha}1}(-p) \delta^4(\theta_1 - \theta_2) , \quad (6.3.23)$$

which can be checked by examining the explicit form of the operators. Another example of the transfer rule is

$$D_1^2 \delta^4(\theta_1 - \theta_2) \bar{D}_2^2 = D_1^2 \bar{D}_1^2 \delta^4(\theta_1 - \theta_2) \neq \bar{D}_1^2 D_1^2 \delta^4(\theta_1 - \theta_2) . \quad (6.3.24)$$

Inside integrals the  $D$ 's can be integrated by parts (see sec. 3.7). Thus

$$\int d^4\theta [D_\alpha(p) f(\theta, p)] g(\theta, -p) = - \int d^4\theta f(\theta, p) D_\alpha(-p) g(\theta, -p) . \quad (6.3.25)$$

This can be most easily understood in  $x$ -space. Since  $D_\alpha = \partial_\alpha + \frac{1}{2} i \bar{\theta}^{\dot{\alpha}} \partial_{\alpha\dot{\alpha}}$  we are doing integration by parts in  $\partial_\alpha$  and in  $\partial_{\alpha\dot{\alpha}}$ .

Armed with these facts we return to the evaluation of the expression in (6.3.21). We concentrate on the  $\theta$  dependence and write the relevant part as

$$\int d^4\theta_1 d^4\theta_2 V(-k, \theta_2) [D_1^2 \bar{D}_1^2 \delta_{12}] [\bar{D}_1^2 D_1^2 \delta_{12}] V(k, \theta_1) . \quad (6.3.26)$$

We have abbreviated  $\delta^4(\theta_1 - \theta_2) = \delta_{12}$ . We now integrate by parts and find first of all

$$\begin{aligned} [D^2 \bar{D}^2 \delta] [\bar{D}^2 D^2 \delta] V &= \bar{D}^2 \delta D^2 [(\bar{D}^2 D^2 \delta) V] \\ &= \delta \bar{D}^2 [(D^2 \bar{D}^2 D^2 \delta) V + (D^\alpha \bar{D}^2 D^2 \delta) D_\alpha V + (\bar{D}^2 D^2 \delta) D^2 V] \\ &= \delta \bar{D}^2 [-p^2 (D^2 \delta) V + p^{\alpha\dot{\alpha}} (\bar{D}_{\dot{\alpha}} D^2 \delta) D_\alpha V + (\bar{D}^2 D^2 \delta) D^2 V] , \end{aligned} \quad (6.3.27)$$

where we have used  $(D)^3 = 0$  and the anticommutation relations  $\{D_\alpha, \bar{D}_{\dot{\alpha}}\} = p_{\alpha\dot{\alpha}}$  when acting on the propagator with momentum  $p$ .

Before proceeding we make the following important observation: Since  $\delta^4(\theta) = \theta^2 \bar{\theta}^2$ , multiplying two identical  $\delta$ -functions together, or multiplying one by  $\theta$  gives zero. We have therefore the following relations:

$$\delta_{21} \delta_{21} = \delta_{21} \delta_{12} = 0 ,$$

$$\delta_{21} D^\alpha \delta_{21} = 0 ,$$

$$\delta_{21} D^2 \delta_{21} = 0 ,$$

$$\delta_{21} D^\alpha \bar{D}^{\dot{\alpha}} \delta_{21} = 0 ,$$

$$\delta_{21} D^\alpha \bar{D}^2 \delta_{21} = 0 ,$$

$$\begin{aligned}\delta_{21} D^2 \bar{D}^2 \delta_{21} &= \delta_{21} \bar{D}^2 D^2 \delta_{21} = \delta_{21} \frac{1}{2} D^\alpha \bar{D}^2 D_\alpha \delta_{21} = \delta_{21} \quad , \\ \delta_{21} D^\alpha \bar{D}^2 D^\beta \delta_{21} &= C^{\beta\alpha} \delta_{21} \quad .\end{aligned}\tag{6.3.28}$$

In these relations, we obtain a nonzero result only if all the  $\theta$ 's in the second  $\delta$ -function are removed by differentiation. Hence two  $D$ 's and two  $\bar{D}$ 's are needed and only their momentum independent parts contribute. Expressions of this kind, but with more  $D$ 's, can be reduced to one of the above forms by using the anticommutation relations. In the expressions with four  $D$ 's the order is irrelevant (except for producing some minus signs).

Returning to our calculation (6.3.27), and letting  $\bar{D}^2$  act on the factors to its right, we see that out of the a priori possible six terms, only three survive:

$$\delta[-p^2(\bar{D}^2 D^2 \delta)V - p^{\alpha\dot{\alpha}}(\bar{D}^{\dot{\beta}} \bar{D}_{\dot{\alpha}} D^2 \delta) \bar{D}_{\dot{\beta}} D_\alpha V + (\bar{D}^2 D^2 \delta)(\bar{D}^2 D^2 V)] \quad .\tag{6.3.29}$$

Finally, using  $\bar{D}^{\dot{\alpha}} \bar{D}_{\dot{\beta}} = \delta_{\dot{\beta}}^{\dot{\alpha}} \bar{D}^2$  we find

$$\delta^4(\theta_1 - \theta_2)[-p^2 - p^{\alpha\dot{\alpha}} \bar{D}_{\dot{\alpha}} D_\alpha + \bar{D}^2 D^2]V(k, \theta_1) \quad .\tag{6.3.30}$$

Inserting this result into the original integral (6.3.21), we use the remaining  $\delta$ -function to do the  $\theta_2$  integral, and finally obtain

$$\frac{1}{2} \int \frac{d^4 k}{(2\pi)^4} d^4 \theta V(-k, \theta) \left[ \int \frac{d^4 p}{(2\pi)^4} \frac{-p^2 - p^{\alpha\dot{\alpha}} \bar{D}_{\dot{\alpha}} D_\alpha + \bar{D}^2 D^2}{p^2(k+p)^2} \right] V(k, \theta).\tag{6.3.31}$$

The result consists of an ordinary loop momentum integral, with usual propagators and some momentum factors in the numerator, and operators  $D, \bar{D}$ , acting on the external superfields (i.e.,  $D$  and  $\bar{D}$  depend on  $k$ , not  $p$ ). The  $p^2$  term is canceled by the tadpole diagram mentioned above, (or gives zero in dimensional regularization) so that the final contribution to the one-loop self-energy is logarithmically divergent. This is a consequence of gauge invariance. The remaining terms in the numerator, in a gauge-invariant regularization (such as dimensional), combine to form  $\frac{1}{2} D^\alpha \bar{D}^2 D_\alpha$ , giving a result proportional to  $W^2$ .

### c. The effective action

In the example above, the  $\delta$ -function has reduced the expression to one which involves a single  $\theta$ . This is a general and important result of superfield perturbation theory. The effective action is a sum of terms involving products of fields evaluated at different points, and a Green function which is a nonlocal function of its arguments:

$$\Gamma = \sum_n \int d^4x_1 \cdots d^4x_n d^4\theta_1 \cdots d^4\theta_n G(x_1 \cdots x_n; \theta_1 \cdots \theta_n) \Phi(x_1, \theta_1) \cdots D^\alpha V(x_i, \theta_i) \cdots \quad (6.3.32)$$

It turns out, however, that by manipulation of the contributions from any graph, we can reduce it to an expression that is local in  $\theta$ , i.e.

$$\Gamma = \sum_n \int d^4x_1 \cdots d^4x_n d^4\theta \tilde{G}(x_1 \cdots x_n) \Phi(x_1, \theta) \cdots D^\alpha V(x_i, \theta) \cdots \quad (6.3.33)$$

We do this as follows: Consider an arbitrary  $L$ -loop contribution to the effective action. It consists of propagators, with factors  $\delta^4(\theta_i - \theta_{i+1})$  and  $D$  operators acting on them, external superfield factors, and  $d^4\theta_i$  integrals. We choose any propagator from a particular vertex  $v$  to another vertex  $v'$ , and integrate by parts to remove all the  $D$ 's from its  $\delta$ -function. The original contribution now becomes a sum of terms. If there are other propagators, each of which connects  $v$  and  $v'$ , we use the relations (6.3.28): The terms vanish unless *each* of the other  $\delta$ -functions has exactly two  $D$ 's and two  $\bar{D}$ 's acting on it, in which case they can be replaced by 1. We now use the free  $\delta$ -function to do the  $\theta$ -integral at  $v'$  and shrink all the propagators between the two vertices to a point in  $\theta$ -space. We repeat the procedure, choosing a propagator leading to a new vertex  $v''$ , until we have removed all  $\delta$ -functions and performed all  $\theta$ -integrals except the original one at  $v$ . Whenever we have more than two  $D$ 's and two  $\bar{D}$ 's on a line we use the anti-commutation relations to replace  $D, \bar{D}$  pairs by momenta. We are left with a sum of terms, all with a single  $\theta$  integral, and various factors of loop-momenta coming from the anticommutators of  $D$ 's, as well as  $D$  factors acting on the external superfields, coming from the integration by parts.

In the course of evaluating Feynman diagrams, we may encounter loop-momentum ultraviolet divergences, and a suitable regularization procedure is needed to handle them. We discuss regularization issues later on. For the time being we assume that

there exists a procedure that allows us to carry out the manipulations we have described above inside momentum integrals.

The expression for the effective action in (6.3.33) reveals one important fact: We have ended up with a  $d^4\theta$  integral, even though in the original classical action we may have had  $d^2\theta$  integrals. This is a consequence of our Feynman rules: All our vertices carry  $d^4\theta$  integrals, and nowhere in our manipulations does a  $d^2\theta$  appear. In particular, if the original action had purely chiral  $d^2\theta$  mass or cubic interaction terms  $\Phi^2$  or  $\Phi^3$ , radiative corrections do not induce finite or infinite modifications of these terms. This is the *no-renormalization theorem* for chiral superfields. Masses and coupling constants *are* renormalized, but only as a consequence of wave function renormalization. (Any  $d^4\theta$  integral can be written as a  $d^2\theta$  integral and a  $\bar{D}^2$  operator acting on the integrand; however, this will not produce the above terms.) This theorem is valid in perturbation theory. So far no one has succeeded in giving examples, in four dimensions, where it might fail nonperturbatively, but a proof of its general validity does not exist. Even within perturbation theory there exists the possibility of a pathological infrared-type behavior which might invalidate it. For example, if in the course of evaluating the effective action a term  $\int d^4\theta \Phi^2 \frac{D^2}{\square} \Phi$  were produced, the  $\bar{D}^2$  operator which comes from converting the integral to chiral form, when acting on the chiral field, would give  $\bar{D}^2 D^2 \square^{-1} \Phi = \Phi$  and we would end up with a contribution to the chiral cubic vertex. Whether such pathological behavior can be obtained in any calculation with a sensible infrared regularization is doubtful.

#### d. Divergences

We now discuss the divergence structure of the effective action. There are two issues involved: We must determine which terms in the effective action are divergent (power counting), and which terms in the classical action lead only to divergences that can be absorbed in a renormalization of the parameters (renormalizable interactions). We restrict our discussion to interacting gauge and chiral scalar superfields (with no negative-dimension coupling constants).

The possible divergences of the effective action can be understood by straight power counting (see sec. 6.6) or simply by a dimensional argument: The divergent parts of graphs that contain no subdivergences give rise to local terms in the effective action of

the form

$$\Gamma_\infty = \int d^4x d^4\theta IP(\Phi, \bar{\Phi}, V, D_\alpha \Phi, \dots) \quad , \quad (6.3.34)$$

where  $IP$  is a polynomial in the fields and their derivatives. Since the effective action must be dimensionless, and  $d^4\theta$  has dimension 2,  $IP$  must also have dimension 2.  $\Phi$  has dimension 1,  $D_\alpha$  has dimension  $\frac{1}{2}$ , and  $V$  is dimensionless. Therefore, graphs with more than two external  $\Phi$ 's are convergent. A  $\Phi\Phi$  or  $\bar{\Phi}\bar{\Phi}$  propagator produces a numerator factor of  $m$  which contributes to the dimension of  $IP$  and therefore reduces the degree of divergence. If  $IP$  is made up of only chiral superfields the  $\theta$  integration will give zero unless some  $D$ 's (at least two of them, to contract indices) are present to make the integrand nonchiral, and again the  $D$ 's contribute to the dimension of  $IP$  reducing the number of fields that can appear. Finally, if gauge invariance requires  $V$  to appear through its field strength  $W_\alpha = i\bar{D}^2 D_\alpha V$  (or its nonabelian generalization), this limits the possible divergences involving  $V$  fields. (This is an oversimplification: We must use the full machinery of Slavnov-Taylor identities, at least in the nonabelian case, or use the background-field method (see sec. 6.5) to analyze the divergences involving gauge superfields.)

The net result of the analysis is to establish that the only local divergent terms contain at most one  $\Phi$  and one  $\bar{\Phi}$  and, while they contain an arbitrary number of  $V$  factors, these enter in a manner which is controlled by the Slavnov-Taylor identities. For a renormalizable theory of chiral scalar multiplets interacting with a vector multiplet (we omit the ghost terms) the renormalized classical action has the form

$$\begin{aligned} & \int d^4x d^4\theta [\bar{\Phi}_R e^{g_R V_R} \Phi_R + tr \nu_R V_R - tr \alpha_R^{-1} (\bar{D}^2 V_R)(D^2 V_R)] \\ & + \frac{1}{g_R^2} \int d^4x d^2\theta tr W_R^2 + [\int d^4x d^2\theta IP(\Phi_R) + h.c.] \quad , \end{aligned} \quad (6.3.35)$$

where the subscript  $R$  labels renormalized quantities. (In the exponential we have written explicitly the gauge coupling constant  $g$  that we normally absorb into  $V$ .)

Since  $V$  is dimensionless,  $V_R$  is in general a nonlinear function of  $V$ , i.e., the wave-function renormalization factor may be a function of  $V$ : We can have functional renormalizations  $V_R = f(V)$ , where each coefficient in the Taylor expansion of  $f$  is a

renormalization constant. Since all such renormalizations are proportional to the field equations ( $\delta S = \int \frac{\delta S}{\delta V} \delta V$ ), they vanish on shell. (Such noncovariant renormalizations are avoided in the background field gauges that we discuss below.)

Ghosts are described by chiral superfields which follow the same rules. The divergences of the theory are all logarithmic, except that of the Fayet-Iliopoulos term, which is quadratic. (However, as we shall discuss in sec. 6.5, this term is not produced by radiative corrections.)

In general, renormalizable interactions are associated with dimensionless (or positive dimension) coupling constants. For Feynman graphs, since at each vertex we have a  $d^4\theta$  integral with dimension 2 and a  $d^4x$  integral with dimension  $-4$ , we may allow up to the equivalent of four  $D$ 's at each vertex. This is indeed the case with the gauge field self-couplings, and also the usual vertices involving chiral superfields, where the  $D$  factors come from our Feynman rules. On the other hand a term such as  $\bar{\Phi}^2\Phi$ , or  $\Phi^4$ , would lead to an excess of  $D$ 's at the vertices and a nonrenormalizable theory.

### e. D-algebra

In the next section we give a number of examples of evaluation of supergraphs. As preparation we discuss several simplifications that we use in performing the manipulation of the  $D$ 's and the  $\theta$  integration. The numerous integrations by parts that have to be performed can lead to long intermediate expressions, and a lot of effort (and paper) can be saved by doing the manipulations directly on the graphs.

We draw the supergraph and indicate on it, adjacent to the vertices, the  $D$  factors acting on the propagators in the order in which they act. We ignore signs having to do with the ordering of the  $D$ 's: These will be determined later. Thus, an expression such as  $\bar{D}^2 D_\alpha \delta^4(\theta - \theta') \bar{D}_\beta \bar{D}_\beta \bar{D}_\gamma$ , with the last three  $D$ 's acting backwards on the  $\theta'$  argument of the  $\delta$ -function (and thus in the order  $D_\beta$  first, then  $\bar{D}_\beta$  next, etc.), would be represented on the graph as shown in fig. 6.3.2:

$$\frac{\bar{D}^2 D_\alpha \quad D_\beta \bar{D}_\beta \cdot D_\gamma}{\hline}$$

Fig. 6.3.2

The transfer rule can be implemented by “sliding” the  $D$ ’s to the left, keeping the order. This corresponds to writing  $-\bar{D}^2 D_\alpha D_\beta \bar{D}_\beta \cdot D_\gamma \delta^4(\theta - \theta')$  with  $D_\gamma$  acting first on the  $\theta$  argument of the  $\delta$ -function. We must keep track of the  $-$  sign coming from transferring an odd number of  $D$ ’s. (Note the order in these expressions, e.g.,  $D_1^\alpha \delta_{12} \bar{D}_2^\beta = D_1^\alpha D_2^\beta \delta_{12} = -D_1^\alpha D_1^\beta \delta_{12}$ .)

We use the commutation relations to replace the rightmost  $\bar{D}, D$  by  $p_{\beta\gamma} \cdot$ . (Since  $p$  is hermitian  $p_{\beta\gamma} = p_{\gamma\beta}$  but we maintain the distinction to keep track of the order in which the  $D$ ’s appeared.) When we encounter expressions such as  $D^2 \bar{D}^2 D^2$  we replace them with  $-p^2 D^2$ .

The integration by parts can also be carried out directly on the graphs. For example, we show in fig. 6.3.3 the integration by parts on a vertex coming from the  $\bar{\Phi} V \Phi$  interaction:

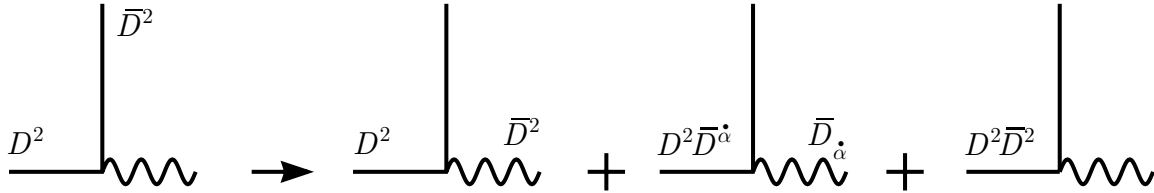


Fig. 6.3.3

Starting from a given graph in general we obtain several, because integration by parts gives several contributions. We remove the  $D$ ’s from any given line and use the  $\delta$ -function to do one of the  $\theta$  integrals, thus contracting the line to a point in  $\theta$  space. We need not indicate explicitly this contraction: A line without any operators on it is understood to be contracted. Whenever several lines connect the same pair of vertices, if all the lines (other than the one we have cleared of  $D$ ’s) have exactly two  $D$ ’s and two  $\bar{D}$ ’s each, we use (6.3.28) to replace them by 1. If any line has fewer  $D$ ’s or  $\bar{D}$ ’s, the contribution vanishes. If any line has more  $D$ ’s or  $\bar{D}$ ’s, we use the anticommutation

relations to reduce their number. In the end we have a sum of graphs, with momentum factors from the anticommutators, and  $D$ 's acting on the external lines only.

To make this procedure clear, we redo the example considered above (fig. 6.3.1, (6.3.21-31)), working directly on the graph, as shown in fig. 6.3.4:

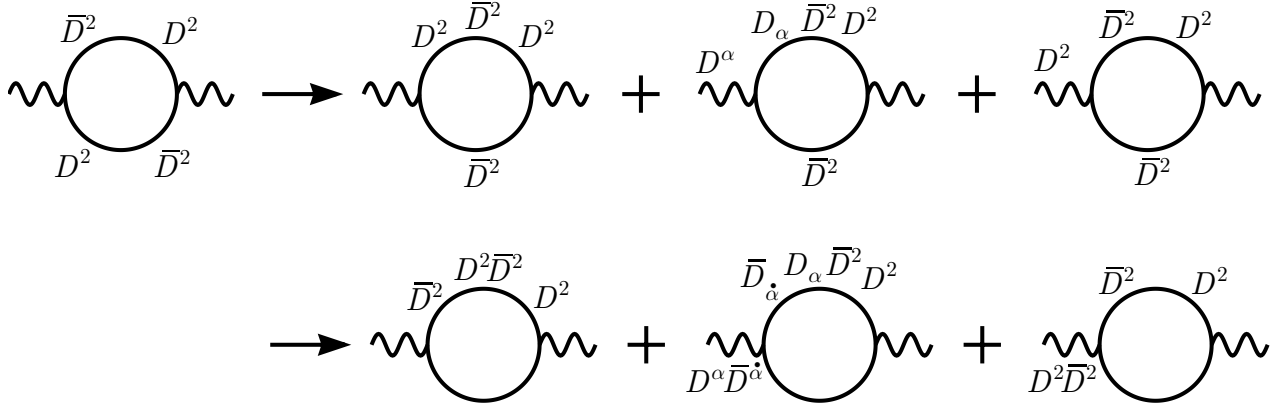


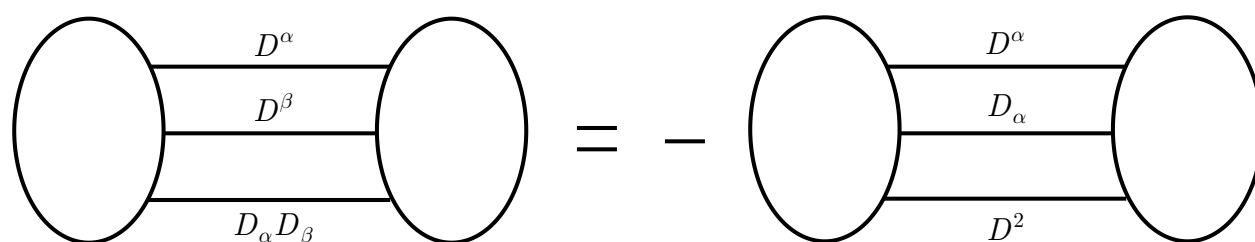
Fig. 6.3.4

In the last step we have only indicated nonzero contributions (the others vanish trivially because of (6.3.28).)

In the course of the manipulations on the graphs, we must keep track of  $-$  signs coming from transfers and from integration by parts. However, we need not keep track of  $-$  signs that come from passing a  $D$  past another  $D$ , nor from signs that come from raising or lowering indices. (On the graphs, we do not indicate the relative order of  $D$ 's on different lines, nor which indices are up or down). These signs can be determined at the end of the computation in the following manner: On the original graph, we have factors such as  $D^2 \bar{D}^2 = \frac{1}{4} D^\alpha D_\alpha \bar{D}^{\dot{\alpha}} \bar{D}_{\dot{\alpha}}$ , and also, from a vertex such as  $V(D^\alpha V)(\bar{D}^2 D_\alpha V)$ , adjacent factors  $D^\alpha$  and  $\frac{1}{2} \bar{D}^{\dot{\alpha}} \bar{D}_{\dot{\alpha}} D_\alpha$  where we determine the initial sign by requiring that in any contracted pair, the first  $D$  or  $\bar{D}$  has the upper index. These various factors may end up in a different order in the final expression, e.g.,  $D^\alpha \dots \bar{D}^{\dot{\alpha}} \dots D_\alpha \dots \bar{D}_{\dot{\alpha}}$ , possibly acting on different superfields; however, we still write a contracted pair with the first index raised. To determine the final overall sign, we count the number of transpositions needed in the final expression to bring the  $D$ 's back to the original order. This is true even if some of the  $D$ 's have been replaced by momenta: An expression such as  $p_{\beta\gamma}$  will

correctly keep track of the transpositions. We do *not* count transpositions of contracted pairs, since the convention for raising and lowering indices cancels such signs:  $X^\alpha Y_\alpha = +Y^\alpha X_\alpha$ . A quick way to count the transpositions is to draw lines connecting all contracted pairs with lines, and count the number of intersections: an odd number means an odd number of transpositions, and hence a  $-$  sign, whereas an even number means no  $-$  sign.

There are many other tricks that one can use to simplify the manipulations. We give the following “twingling” rule which is often useful:



*Fig. 6.3.5*

We are now ready to consider further examples of graph evaluation.

## 6.5. The background field method

### a. Ordinary Yang-Mills

The background field method is extremely useful in supersymmetric Yang-Mills theory, and essential in the quantum theory of supergravity. In this section we review the background field method for ordinary Yang-Mills, and then extend it to the supersymmetric case. The extension involves some subtleties, primarily because of the nonlinearity of the gauge transformations.

In gauge theories we start with the gauge-fixed functional integral (6.2.9) and introduce sources coupled to the fields, defining

$$Z(J) = \int \mathcal{D}A \mathcal{D}c \mathcal{D}c' e^{S_{\text{eff}} + JA} \quad , \quad JA \equiv \int d^4x J^a A_a \quad . \quad (6.5.1)$$

We introduce  $W(J) = \ln Z(J)$  and define the effective action by a Legendre transform

$$\Gamma(\hat{A}) = W(J) - J\hat{A} \quad , \quad \hat{A}_a = \frac{\delta W}{\delta J^a} \quad . \quad (6.5.2)$$

This quantity is not gauge invariant in general. Physical quantities computed from it are gauge invariant, and the Green functions satisfy Slavnov-Taylor identities that express the underlying gauge invariance, but manifest gauge invariance is lost because of the gauge fixing procedure. On the other hand, the effective action computed in the background field method is manifestly gauge invariant. It is equivalent to the usual one, but is more convenient to handle.

In the background field quantization of Yang-Mills theories we follow a procedure similar to that of sec. 6.2a. We start with the gauge-invariant Lagrangian  $\mathcal{L}_{\text{inv}}(A_a)$  (other fields may be present but we do not indicate them explicitly) and *split* the field into a background and quantum part:  $\mathcal{L}_{\text{inv}}(\mathbf{A}_a + A_a)$ . The action is invariant under two kinds of transformations that give the same  $\delta(\mathbf{A}_a + A_a)$ :

Quantum:

$$\delta \mathbf{A}_a = 0 \quad , \quad \delta A_a = \nabla_a \lambda + i[\lambda, A_a] \quad , \quad (6.5.3)$$

$$\nabla_a \lambda = \partial_a \lambda + i[\lambda, \mathbf{A}_a] \quad ,$$

Background:

$$\delta \mathbf{A}_{\underline{a}} = \nabla_{\underline{a}} \lambda \quad , \quad \delta A_{\underline{a}} = i[\lambda, A_{\underline{a}}] \quad . \quad (6.5.4)$$

We consider now the functional

$$\mathbf{Z}(\mathbf{A}) = \int \mathcal{D}A_{\underline{a}} e^{\int \mathcal{L}_{inv}(A_{\underline{a}} + \mathbf{A}_{\underline{a}})} \quad , \quad (6.5.5)$$

and quantize as before to fix the *quantum* gauge invariance except that, to maintain manifest invariance with respect to the *background* gauge transformations, we choose the gauge-fixing function so that it transforms *covariantly* under these transformations. This requires in particular that we covariantize the derivatives that appear there with respect to the background field:  $\partial^{\underline{a}} A_{\underline{a}} \rightarrow \nabla^{\underline{a}} A_{\underline{a}} = \partial^{\underline{a}} A_{\underline{a}} - i[\mathbf{A}^{\underline{a}}, A_{\underline{a}}]$ . The remainder of the quantization procedure is the same. We require the Faddeev-Popov ghosts to transform covariantly under background gauge transformations, and we choose the weighting function  $\exp(-\frac{1}{\alpha g^2} \text{tr} \int f^2)$  to be invariant. We thus obtain the following expression :

$$\mathbf{Z}(\mathbf{A}) = \int \mathcal{D}A_{\underline{a}} \mathcal{D}c \mathcal{D}c' e^{S_{inv}(A + \mathbf{A}) - \int \frac{1}{4\alpha g^2} \text{tr}(\nabla \cdot A)^2 + S_{FP}} \quad . \quad (6.5.6)$$

$\mathbf{Z}$  is manifestly invariant under background gauge transformations but its significance is not obvious. To elucidate its meaning we consider an object defined exactly like  $\mathbf{Z}$  except that we also couple the quantum field to a source:

$$\tilde{Z}(J, \mathbf{A}) = \int \mathcal{D}A_{\underline{a}} \mathcal{D}c \mathcal{D}c' e^{S_{eff}(A_{\underline{a}}, \mathbf{A}_{\underline{a}}) + JA} \quad . \quad (6.5.7)$$

We can now pass from  $\tilde{Z}(J, \mathbf{A})$  to  $\tilde{\Gamma}(\tilde{A}, \mathbf{A})$  by a Legendre transformation in the presence of the fixed field  $\mathbf{A}$ . On the other hand, returning to  $\tilde{Z}$  itself, we can make a change of variables  $A \rightarrow A - \mathbf{A}$  which gives

$$\begin{aligned} \tilde{Z}(J, \mathbf{A}) &= e^{-J\mathbf{A}} \int \mathcal{D}A_{\underline{a}} \mathcal{D}c \mathcal{D}c' e^{\int [\mathcal{L}_{inv}(A_{\underline{a}}) + \mathcal{L}_{GF} + \mathcal{L}_{FP} + JA]} \\ &= e^{-J\mathbf{A}} Z(J, \mathbf{A}) \quad . \end{aligned} \quad (6.5.8)$$

It contains the usual  $\mathcal{L}_{inv}(A_{\underline{a}})$  but unusual gauge fixing and ghost terms that have additional dependence on  $\mathbf{A}_{\underline{a}}$ . Therefore,  $Z(J, \mathbf{A})$  is the usual generating functional but

with  $\mathbf{A}$ -dependent gauge fixing and ghost terms. Now we have

$$\tilde{W}(J, \mathbf{A}) = \ln \tilde{Z}(J, \mathbf{A}) = \ln Z(J, \mathbf{A}) - J\mathbf{A} = W(J, \mathbf{A}) - J\mathbf{A} \quad , \quad (6.5.9)$$

and

$$\begin{aligned} \tilde{\Gamma}(\tilde{A}, \mathbf{A}) &= \tilde{W}(J, \mathbf{A}) - J\tilde{A} = [W(J, \mathbf{A}) - J\mathbf{A}] - J\tilde{A} \\ &= W(J, \mathbf{A}) - J(\tilde{A} + \mathbf{A}) \quad , \quad \tilde{A} = \frac{\delta \tilde{W}}{\delta J} = \frac{\delta W}{\delta J} - \mathbf{A} \quad . \end{aligned} \quad (6.5.10)$$

Therefore

$$\tilde{\Gamma}(\tilde{A}, \mathbf{A}) = \Gamma(\tilde{A} + \mathbf{A}, \mathbf{A}) \quad , \quad (6.5.11)$$

is the usual effective action  $\Gamma(\hat{A})$ , evaluated in an unusual  $\mathbf{A}$ -dependent gauge, and at  $\hat{A} = \tilde{A} + \mathbf{A}$ . In particular, if in the evaluation of  $\tilde{\Gamma}(\tilde{A}, \mathbf{A})$  we restrict ourselves to graphs with no external  $\tilde{A}$  lines (“vacuum” graphs), i.e., set  $\tilde{A} = 0$ , we will obtain  $\Gamma(\mathbf{A})$ , the usual effective action. But these “ $\tilde{A}$ -vacuum” graphs are simply the one-particle-irreducible subset of the graphs obtained from  $\tilde{Z}(0, \mathbf{A}) = \mathbf{Z}(\mathbf{A})$ . (Actually, this is an oversimplification. What one obtains is not exactly the effective action, because  $\mathbf{A}$  lines from the gauge-fixing term give additional contributions. However, because of gauge invariance, it can be shown that these have no effect on S-matrix elements so that the identification, though strictly speaking not correct, can be used when computing physical quantities.)

Our conclusion is that the effective action is obtained from  $\mathbf{Z}(\mathbf{A})$  by evaluating in perturbation theory one-particle-irreducible graphs with only internal  $A_a$  lines and external  $\mathbf{A}_a$  lines (as well as ghost, and other non-gauge field lines). In particular, if we expand  $\mathcal{L}_{inv}(\mathbf{A} + A) = \mathcal{L}(\mathbf{A}) + \mathcal{L}'(\mathbf{A})A + \mathcal{L}''(\mathbf{A})A^2 + \dots$ , the first term does not contribute to loop graphs (it is the classical contribution to  $\Gamma$ ), and the second can be dropped because it does not contribute to one-particle-irreducible graphs with no external  $A$  lines. The  $A^2$  term gives the complete contribution (from the gauge field) to one-loop graphs. For higher-loop graphs, internal vertices are read from the higher order expansion, and all the terms contribute to vertices that involve the external  $\mathbf{A}$  lines.

There is an additional feature of the background-field quantization that is not usually encountered in the Yang-Mills case but that is important. This is the appearance of the Nielsen-Kallosh ghost. In the gauge-averaging procedure we used the simplest exponential factor to produce the gauge-fixing term in the effective Lagrangian. However, a more complicated averaging function could be used, e.g.,  $\exp \int (fMf)$  where  $M$  is any operator (matrix). To properly normalize the averaging procedure, we must divide by  $\det M$ . If  $M$  is field independent, this is a trivial factor. However, if  $M$  is a function of the background field, we normalize the gauge averaging by introducing into the functional integral a factor

$$\int \mathcal{D}f \mathcal{D}b e^{fMf} e^{bMb} , \quad (6.5.12)$$

where  $b$  is a *ghost* field, with opposite statistics to  $f$ . When we carry out the  $f$  integration using the  $\delta$ -function of sec. 6.2.a, we are left with the  $b$  field. Thus, the final form is

$$\mathbf{Z}(\mathbf{A}_{\underline{a}}) = \int \mathcal{D}A_{\underline{a}} \mathcal{D}c \mathcal{D}c' \mathcal{D}b e^{\int [\mathcal{L}_{inv}(A + \mathbf{A}) + \mathcal{L}_{GF}(A, \mathbf{A}) + \mathcal{L}_{FP}(c, c', A, \mathbf{A}) + \mathcal{L}_{NK}(b, \mathbf{A})]} \quad (6.5.13)$$

where  $\mathcal{L}_{NK} = bMb$ . If  $M$  is independent of the background field, the additional ghost gives trivial contributions and can be dropped; otherwise, since the ghost field  $b$  has no interactions with other quantum fields and since it enters quadratically, it only contributes at the one-loop level.

To motivate the procedure we use in the background field quantization of supersymmetric Yang-Mills theory, we point out two aspects of the background-quantum splitting  $A_{\underline{a}} \rightarrow \mathbf{A}_{\underline{a}} + A_{\underline{a}}$  of ordinary Yang-Mills. This splitting has the virtue that the transformations  $\delta(\mathbf{A}_{\underline{a}} + A_{\underline{a}}) = (\partial_{\underline{a}}\omega - i[\mathbf{A}_{\underline{a}}, \omega]) + (-i[A_{\underline{a}}, \omega])$ , which leave the action invariant, can be interpreted as ordinary gauge transformations of the background field accompanied by covariant gauge rotations (linear and homogeneous) of the quantum field. Furthermore, in an expansion

$$\mathcal{L}(A + \mathbf{A}) = \sum [\mathcal{L}^{(n)}(\mathbf{A})](A)^n , \quad (6.5.14)$$

each term in the power series is separately invariant under these transformations, since  $A_{\underline{a}}$  transforms linearly and homogeneously, as do the functional derivatives of  $\mathcal{L}(\mathbf{A})$ .

Thus, if we truncate the series, as we do in a perturbative loop-by-loop evaluation of the effective action, we maintain the background gauge invariance. *This would not be true if the transformation of the quantum field were nonlinear.*

## b. Supersymmetric Yang-Mills

In supersymmetric Yang-Mills theory the classical action is invariant under *non-linear* gauge transformations  $e^V \rightarrow e^{V'} = e^{i\bar{\Lambda}} e^V e^{-i\Lambda}$ , and the splitting  $V \rightarrow V + \mathbf{V}$  is unsuitable. To motivate the subsequent procedure, we first reexamine the background-quantum splitting of ordinary Yang-Mills theory from a different point of view. We start with the original gauge and matter action, invariant under the local transformations  $\delta A_a = \partial_a \omega - i[A_a, \omega]$  and, for some matter field,  $\delta\psi = i[\omega, \psi]$ . Under *global* transformations with constant  $\omega$  we still have invariance, with the gauge fields rotating like the matter fields. For local  $\omega$ , we can introduce a new invariance by keeping the covariant transformations  $i[\omega, A_a]$  and  $i[\omega, \psi]$  for all fields, and introducing a separate gauge field, the background field  $\mathbf{A}_a$  to covariantize the derivatives. Since in the original action all derivatives entered in the form  $\nabla_a = \partial_a - i[A_a, \ ]$ , this covariantization amounts to the replacement

$$\nabla_a = \partial_a - i[A_a, \ ] \rightarrow \nabla_a - i[A_a, \ ] = \partial_a - i[\mathbf{A}_a + A_a, \ ] \quad , \quad (6.5.15)$$

which is equivalent to the ordinary quantum-background splitting. We now have two invariances: the original one where the background field is inert, and the new one, under which all the fields transform. *We obtain a linear splitting  $A \rightarrow A + \mathbf{A}$  because the gauge field enters linearly in the covariant derivative.* In supersymmetric Yang-Mills theory this is so in the abelian case, but not in the nonabelian case. However, the philosophy is the same. We start with the locally invariant gauge theory, observe that it is invariant for global transformations (with  $\Lambda = \lambda = \bar{\Lambda}$  a constant matrix, not a superfield), under which the gauge fields transform covariantly (linearly and homogeneously, since  $e^V \rightarrow e^{i\lambda} e^V e^{-i\lambda}$  implies  $V \rightarrow e^{i\lambda} V e^{-i\lambda}$ ) and now gauge this transformation by covariantizing with the aid of a background field. This amounts to the replacement  $D_A \rightarrow \nabla_A$  where  $\nabla_A$  is a background covariant derivative. We also have to treat the covariantly chiral superfields properly.

We recall that supersymmetric Yang-Mills theory can be formulated in terms of constrained covariant derivatives. The reason for solving the constraints and introducing

the gauge prepotentials is that only these unconstrained objects are suitable for quantization. In solving the constraints we have the choice of working in the vector representation or in the chiral representation. The latter is more convenient for quantization, expressing the theory in terms of the real superfield  $V$ , rather than superfields  $\Omega, \bar{\Omega}$  with a redundant gauge invariance. We want to maintain this advantage in the background field method and work with a quantum  $V$ . On the other hand, when we introduce the background covariant derivatives, it is useful to think of them in the vector representation. In fact it is possible to express all our results in terms of the (constrained) background covariant derivatives themselves, without ever introducing explicitly the background gauge superfields, i.e. without solving the constraints, and in that case the only representation that is available is the vector representation. The advantage of working with the background derivatives directly is that background covariance is manifest and we obtain significant simplifications and improvement in the power counting rules for Feynman graphs.

We also express covariantly chiral superfields in terms of the quantum field  $V$  and background-covariantly chiral superfields. The latter therefore depend implicitly on the background fields and would seem not to be suitable for quantization. However, this is not always the case: At more than one loop, and even at one loop for real representations of the gauge group, we formulate covariant Feynman rules directly for covariantly chiral superfields that lead to considerable improvement over the ordinary ones.

Starting with the ordinary covariant derivatives we perform the splitting by writing them, in the *quantum-chiral but background-vector* representation, as

$$\nabla_\alpha = e^{-V} \bar{\nabla}_\alpha e^V, \quad \bar{\nabla}_{\dot{\alpha}} = \bar{\nabla}_{\dot{\alpha}}, \quad \nabla_{\underline{a}} = -i\{\nabla_\alpha, \bar{\nabla}_{\dot{\alpha}}\} \quad ; \quad (6.5.16)$$

where  $\bar{\nabla}_\alpha$  and  $\bar{\nabla}_{\dot{\alpha}}$  are background covariant derivatives satisfying the usual constraints. The  $\nabla$ 's transform covariantly under two sets of transformations:

(a) Quantum:

$$e^V \rightarrow e^{i\bar{\Lambda}} e^V e^{-i\Lambda},$$

$$\bar{\nabla}_A \rightarrow \bar{\nabla}_A, \quad (6.5.17)$$

with background covariantly chiral parameters  $\bar{\nabla}_\alpha \bar{\Lambda} = \bar{\nabla}_{\dot{\alpha}} \Lambda = 0$ , i.e.,

$$\nabla_A \rightarrow e^{i\Lambda} \nabla_A e^{-i\Lambda} . \quad (6.5.18)$$

(b) Background:

$$\begin{aligned} e^V &\rightarrow e^{iK} e^V e^{-iK} , \\ \nabla_A &\rightarrow e^{iK} \nabla_A e^{-iK} , \end{aligned} \quad (6.5.19)$$

with a real parameter  $K = \bar{K}$ , i.e.,

$$\nabla_A \rightarrow e^{iK} \nabla_A e^{-iK} . \quad (6.5.20)$$

The background field transformations of  $V$  can be rewritten as

$$V \rightarrow e^{iK} V e^{-iK} , \quad (6.5.21)$$

i.e.,  $V$  transforms covariantly.

\* \* \*

While this procedure has given us a correct quantum-background splitting, in contrast to the component Yang-Mills case it results in different transformations of  $\nabla_A$  under quantum and background transformations. However, the transformation of the unsplit gauge field *is* the same. To understand the splitting of the gauge field we solve the constraints on the background covariant derivatives:

$$\nabla_\alpha = e^{-\Omega} D_\alpha e^\Omega , \quad \bar{\nabla}_{\dot{\alpha}} = e^{\bar{\Omega}} \bar{D}_{\dot{\alpha}} e^{-\bar{\Omega}} . \quad (6.5.22)$$

Hence the splitting of the full derivatives is

$$\nabla_\alpha = e^{-V} e^{-\Omega} D_\alpha e^\Omega e^V , \quad \bar{\nabla}_{\dot{\alpha}} = e^{\bar{\Omega}} \bar{D}_{\dot{\alpha}} e^{-\bar{\Omega}} . \quad (6.5.23)$$

We transform to a background chiral representation by pre- and post-multiplying all quantities by  $e^{-\bar{\Omega}}$  and  $e^{\bar{\Omega}}$ , respectively. Then

$$\nabla_\alpha \rightarrow e^{-\bar{\Omega}} e^{-V} e^{-\Omega} D_\alpha e^\Omega e^V e^{\bar{\Omega}} , \quad \bar{\nabla}_{\dot{\alpha}} \rightarrow \bar{D}_{\dot{\alpha}} , \quad (6.5.24)$$

and the splitting is equivalent to replacing  $e^V$  by

$$e^{V(split)} = e^\Omega e^V e^{\bar{\Omega}} . \quad (6.5.25)$$

In other words, we split the full  $V$  into a quantum  $V$  and background  $\Omega$  and  $\bar{\Omega}$  in a

particular, nonlinear fashion. (In the abelian case this reduces to  $V \rightarrow V + \mathbf{\Omega} + \bar{\mathbf{\Omega}}$  which is just the ordinary splitting since by (4.2.72)  $\mathbf{\Omega} + \bar{\mathbf{\Omega}} = \mathbf{V}$ .)

The usual chiral representation transformations of (6.5.25)

$$(e^{\mathbf{\Omega}} e^V e^{\bar{\mathbf{\Omega}}})' = e^{i\bar{\Lambda}_0} (e^{\mathbf{\Omega}} e^V e^{\bar{\mathbf{\Omega}}}) e^{-i\Lambda_0} \quad , \quad (6.5.26)$$

(where  $\Lambda_0$  is ordinary chiral,  $\bar{D}_{\dot{\alpha}} \Lambda_0 = 0$ ), can be written in two ways:

(a)

$$\begin{aligned} (e^{\mathbf{\Omega}} e^V e^{\bar{\mathbf{\Omega}}})' &= e^{\mathbf{\Omega}} [(e^{-\mathbf{\Omega}} e^{i\bar{\Lambda}_0} e^{\mathbf{\Omega}}) e^V (e^{\bar{\mathbf{\Omega}}} e^{-i\Lambda_0} e^{-\bar{\mathbf{\Omega}}})] e^{\bar{\mathbf{\Omega}}} \\ &= e^{\mathbf{\Omega}} (e^{i\bar{\Lambda}} e^V e^{-i\Lambda}) e^{\bar{\mathbf{\Omega}}} \quad , \end{aligned} \quad (6.5.27a)$$

i.e., the quantum transformations (6.5.17), *with background covariantly chiral*  $\Lambda$ , or

(b)

$$(e^{\mathbf{\Omega}} e^V e^{\bar{\mathbf{\Omega}}})' = (e^{i\bar{\Lambda}_0} e^{\mathbf{\Omega}} e^{-iK}) (e^{iK} e^V e^{-iK}) (e^{iK} e^{\bar{\mathbf{\Omega}}} e^{-i\Lambda_0}) \quad , \quad (6.5.27b)$$

i.e., the background transformations (6.5.19) (cf. (4.2.70-71); recall that the  $\Lambda_0$  part of the transformation of  $\mathbf{\Omega}$  does not affect the transformation of the background covariant derivatives). This is very similar to the situation in component Yang-Mills.

The gauge Lagrangian has the form

$$tr W^2 = - tr \left( \frac{1}{2} [\bar{\nabla}^{\dot{\alpha}}, \{\bar{\nabla}_{\dot{\alpha}}, \nabla_{\alpha}\}] \right)^2 \quad . \quad (6.5.28)$$

When we substitute (6.5.16) into (6.5.28), we obtain a splitting of the action into explicit quantum  $V$ 's and background covariant derivatives. Since the  $\nabla$ 's transform covariantly, the Lagrangian will be invariant under both background and quantum transformations. Furthermore, since  $V$  transforms homogeneously, expanding the Lagrangian in powers of  $V$  will maintain the background invariance term-by-term, which is one of the required properties of a good splitting.

When covariantly chiral superfields  $\Phi$ ,  $\bar{\nabla}_{\dot{\alpha}} \Phi = 0$  are present, we first express them in terms of background covariantly chiral superfields by  $\Phi = \tilde{\Phi}$ ,  $\tilde{\Phi} = \bar{\Phi} e^V$  (in the *quantum* chiral representation)  $\bar{\nabla}_{\dot{\alpha}} \Phi = \nabla_{\alpha} \bar{\Phi} = 0$ , and then *linearly* split them into a sum of background and quantum fields. The quantum fields transform under

(a) Quantum transformations:

$$\begin{aligned}\Phi' &= e^{i\Lambda}\Phi \quad , \\ \bar{\Phi}' &= \bar{\Phi}e^{-i\bar{\Lambda}} \quad .\end{aligned}\tag{6.5.29}$$

(b) Background transformations:

$$\begin{aligned}\Phi' &= e^{iK}\Phi \quad , \\ \bar{\Phi}' &= \bar{\Phi}e^{-iK} \quad .\end{aligned}\tag{6.5.30}$$

The chiral field action is invariant under both quantum and background transformations.

We examine now the background field quantization. We proceed as in the conventional approach, but compute

$$\mathbf{Z} = \int \mathcal{D}V \mathcal{D}c \mathcal{D}c' \mathcal{D}\bar{c} \mathcal{D}\bar{c}' \delta(\bar{\nabla}^2 V - f) \delta(\nabla^2 V - \bar{f}) e^{S_{inv} + S_{FP}} \quad .\tag{6.5.31}$$

We have chosen background-covariantly chiral gauge fixing functions, and this means that the Faddeev-Popov ghosts, introduced as in sec. 6.2, are also background covariantly chiral. Finally, we gauge average with

$$\int \mathcal{D}f \mathcal{D}\bar{f} \mathcal{D}b \mathcal{D}\bar{b} e^{-\int d^4x d^4\theta [\bar{f}f + \bar{b}b]} \quad ,\tag{6.5.32}$$

where the background covariantly chiral Nielsen-Kallosh ghosts  $b$ ,  $\bar{b}$  have been introduced to normalize to 1 the averaging over  $f$ ,  $\bar{f}$ . This leads to the final form

$$\begin{aligned}\mathbf{Z} &= \int \mathcal{D}V \mathcal{D}c \mathcal{D}c' \mathcal{D}\bar{c} \mathcal{D}\bar{c}' \mathcal{D}b \mathcal{D}\bar{b} e^{S_{eff}} \quad , \\ S_{eff} &= S_{inv} + S_{GF} + S_{FP} + \int \bar{b}b \quad ;\end{aligned}\tag{6.5.33}$$

which, except for the Nielsen-Kallosh ghosts, is like (6.2.19), but with background covariant derivatives and covariantly chiral superfields. The N.-K. ghosts interact with the background field, and only give one-loop contributions. If we couple external sources to the quantum fields, e.g.,  $\int d^4x d^4\theta JV$ , the generating functional  $\mathbf{Z}(J, \mathbf{\Omega})$  will still be invariant under background transformations, provided we require the sources to

transform covariantly,  $\delta J = i[K, J]$ .

What we must do now is argue that the background field functional, obtained by setting sources to zero and computing one-particle-irreducible graphs with only internal quantum lines and external background lines, is equivalent to the usual effective action, except for being computed in a different gauge. This is less direct than in the ordinary case because the splitting is highly nonlinear. We present the following argument:

The splitting (6.5.25) is

$$V \rightarrow V + \mathbf{\Omega} + \bar{\mathbf{\Omega}} + \text{nonlinear terms} . \quad (6.5.34)$$

In a gauge for the background fields where  $\mathbf{\Omega} = \bar{\mathbf{\Omega}} = \frac{1}{2} \mathbf{V}$  we write this as  $V \rightarrow f(V, \mathbf{V})$  where  $f(0, \mathbf{V}) = \mathbf{V}$  and  $f(V, 0) = V$ . If now in the original functional integral we add a source term  $\int d^4x d^4\theta J[f(V, \mathbf{V}) - f(0, \mathbf{V})]$  to define a  $\tilde{Z}(J, \mathbf{V})$  we will have a  $JV$  coupling, and coupling to higher order terms in  $V$  and  $\mathbf{V}$  (which are irrelevant when computing the S-matrix), but no linear coupling  $J\mathbf{V}$ . When we set  $\mathbf{V} = 0$  we obtain the conventional  $Z(J)$ . As in (6.5.8) we make a change of variables of integration which involves the *inverse* of the function  $f$ . Under this transformation the invariant gauge action goes to its usual form in terms of  $V$ , the gauge fixing and ghost terms change in a complicated, but physically irrelevant manner, and the coupling to the source becomes simply  $JV - J\mathbf{V}$ . Furthermore, the Jacobian of the transformation is 1 (see sec. 3.8.b). We now have the same form as in ordinary Yang-Mills theory, and we conclude that the background field functional computed by setting  $J = 0$ , i.e., evaluating graphs with only internal quantum lines, does give the usual effective action as a function of the background field, albeit in an unconventional gauge. Therefore all physically relevant quantum corrections can be obtained from the background field functional. We now discuss how to evaluate it in perturbation theory.

### c. Covariant Feynman rules

We consider first contributions from only the quantum gauge field  $V$ . The effective Lagrangian is

$$-\frac{1}{2g^2} \text{tr}[(e^{-V} \nabla^\alpha e^V) \bar{\nabla}^2 (e^{-V} \nabla_\alpha e^V) + V(\nabla^2 \bar{\nabla}^2 + \bar{\nabla}^2 \nabla^2)V] . \quad (6.5.35)$$

All the dependence on the background fields is through the connection coefficients and

never through the gauge fields themselves.

The quadratic action has the form

$$-\frac{1}{2g^2} \text{tr} V \left[ -\nabla^\alpha \bar{\nabla}^2 \nabla_\alpha - i \mathbf{W}^\alpha \nabla_\alpha + \frac{1}{2} i (\nabla_\alpha \mathbf{W}^\alpha) + \nabla^2 \bar{\nabla}^2 + \bar{\nabla}^2 \nabla^2 \right] V \quad (6.5.36)$$

Using the commutation relations

$$[\bar{\nabla}_{\dot{\alpha}}, \nabla_{\alpha\dot{\beta}}] = C_{\dot{\alpha}\dot{\beta}} \mathbf{W}_\alpha \quad , \quad (6.5.37)$$

this can be rewritten as

$$-\frac{1}{2g^2} \text{tr} V [\mathbf{\square} - i \mathbf{W}^\alpha \nabla_\alpha - i \bar{\mathbf{W}}^{\dot{\alpha}} \bar{\nabla}_{\dot{\alpha}}] V \quad , \quad (6.5.38)$$

where  $\mathbf{\square} = \frac{1}{2} \nabla^a \nabla_a$  is the background covariant d'Alembertian and  $\mathbf{W}_\alpha$  is the background field strength. Introducing connection coefficients (depending on the background fields) by  $\nabla_A = D_A - i \mathbf{\Gamma}_A$ , we can separate out a free kinetic term, and interactions with the background:

$$\begin{aligned} & -\frac{1}{2g^2} \text{tr} V [\mathbf{\square}_0 - i \mathbf{\Gamma}^a \partial_a - \frac{1}{2} i (\partial^a \mathbf{\Gamma}_a) - \frac{1}{2} \mathbf{\Gamma}^a \mathbf{\Gamma}_a \\ & - i \mathbf{W}^\alpha (D_\alpha - i \mathbf{\Gamma}_\alpha) - i \bar{\mathbf{W}}^{\dot{\alpha}} (\bar{D}_{\dot{\alpha}} - i \bar{\mathbf{\Gamma}}_{\dot{\alpha}})] V \quad . \end{aligned} \quad (6.5.39)$$

This expression is sufficient for doing one-loop calculations using conventional propagators for real scalar superfields and the usual  $D$ -manipulations. Since the interaction with the background fields is at most linear in  $D$ 's, and at least four  $D$ 's are needed in a loop, *the first nonvanishing one-loop contribution from  $V$  is in the four-point function.* Self-interactions for computing higher-loop contributions can be obtained from the higher-order in  $V$  terms in the Lagrangian (6.5.35).

We now turn to contributions from (fully) covariantly chiral physical superfields and background covariantly chiral ghost superfields. In principle we have to solve the chirality constraint  $\bar{\nabla}_{\dot{\alpha}} \Phi = 0$  (by writing  $\Phi = e^{\bar{\nabla}} \Phi_0$  in terms of an ordinary chiral superfield), but this introduces explicit dependence on the background gauge prepotentials which we wish to avoid if possible. Instead, we reexamine the derivation of the Feynman rules for chiral superfields.

We consider the generating functional of the form

$$Z(j, \bar{j}) = \int \mathcal{D}\Phi \mathcal{D}\bar{\Phi} e^{S + (\int d^4x d^2\theta j\Phi + h.c.)} , \quad (6.5.40)$$

where  $\Phi$  and  $j$  are covariantly chiral  $\bar{\nabla}_{\dot{\alpha}}\Phi = \bar{\nabla}_{\dot{\alpha}}j = 0$ . For the time being we need not specify whether these are full covariant derivatives or just background covariant derivatives. In principle we define  $\int \mathcal{D}\Phi$  as the integral over the corresponding chiral-representation field  $\Phi_0$  (antichiral for  $\bar{\Phi}$  integration), but in practice we simply define it by the Gaussian integral

$$\int \mathcal{D}\Phi e^{\int d^4x d^2\theta \frac{1}{2}\Phi^2} = 1 \quad . \quad (6.5.41)$$

Additional fields may be present but we need not indicate them explicitly.

We *define* covariant functional differentiation by

$$\frac{\delta\Phi(z)}{\delta\Phi(z')} = \bar{\nabla}^2 \delta^8(z - z') \quad . \quad (6.5.42)$$

This form can be derived from (3.8.3), or by writing  $\Phi = \bar{\nabla}^2\Psi$ , in terms of a general superfield, and covariantizing (3.8.13). Manifestly covariant rules for chiral superfields can now be found by a direct covariantization of the usual method. The covariantization of the identity  $\bar{D}^2 D^2\Phi = \square_0\Phi$  (where  $\square_0$  denotes now the free d'Alembertian) becomes

$$\bar{\nabla}^2 \nabla^2 \Phi = \square_+ \Phi \quad ,$$

$$\square_+ = \square - iW^\alpha \nabla_\alpha - \frac{1}{2}i(\nabla^\alpha W_\alpha) \quad , \quad (6.5.43)$$

with the covariant  $\square$ . We consider first the massless case.

We carry out the functional integration over  $\Phi$  by separating out the interaction terms and doing the Gaussian integral, and obtain

$$Z = \Delta \cdot e^{S_{int}(\frac{\delta}{\delta j}, \frac{\delta}{\delta \bar{j}})} e^{-\int d^4x d^4\theta \bar{j} \square_+^{-1} j} \quad , \quad (6.5.44)$$

where  $\Delta$  is the functional determinant

$$\Delta = \int \mathcal{D}\Phi \mathcal{D}\bar{\Phi} e^{S_0} \quad , \quad S_0 = \int d^4x d^4\theta \bar{\Phi} \Phi \quad . \quad (6.5.45)$$

In general the above expression must still be integrated over other quantum fields that may be present.

Before we evaluate  $\Delta$ , which will give a separate, one-loop contribution to the effective action from the  $\Phi$  field, we examine the rest of the contributions. The expression for  $Z$  is identical to the one in (6.3.14), except for the presence of covariant derivatives and covariantly chiral sources, and the factor  $\Delta$ . The perturbation expansion takes the same form, except that from the functional differentiation we get factors of  $\bar{\nabla}^2$  or  $\nabla^2$  acting on chiral and antichiral lines. The propagators are given by  $-\square_+^{-1}$ , but in a perturbative calculation we separate  $\square_+$  into a free part, which leads to  $p^{-2}$  propagators, and the remainder, which gives additional interaction vertices. However, at no stage do we encounter explicit gauge fields, only connections and field strengths. (The explicit dependence on the quantum gauge fields will be needed only when we functionally integrate over them.)

We now evaluate  $\Delta$ . It gives the complete one-loop contribution of the chiral superfield to graphs with only external  $V$  lines and could be evaluated by using standard Feynman rules, but this we wish to avoid. This turns out to be possible only for *real* representations of the Yang-Mills group. Of course, real representations are frequently the ones of interest: e.g., the Yang-Mills ghosts are in the adjoint representation, which is always real. We therefore consider first the case of real representations, and return later to the complications caused by complex representations. We are still considering the massless case.

The action  $S_0$  leads to the equations of motion (in the presence of sources)

$$\mathbf{O} \begin{pmatrix} \Phi \\ \bar{\Phi} \end{pmatrix} + \begin{pmatrix} j \\ \bar{j} \end{pmatrix} = 0 \quad , \quad \mathbf{O} \equiv \begin{pmatrix} 0 & \bar{\nabla}^2 \\ \nabla^2 & 0 \end{pmatrix} \quad . \quad (6.5.46)$$

We define an action whose equations of motion are

$$\mathbf{O}^2 \begin{pmatrix} \Phi \\ \bar{\Phi} \end{pmatrix} - \begin{pmatrix} j \\ \bar{j} \end{pmatrix} = 0 \quad , \quad \mathbf{O}^2 = \begin{pmatrix} \bar{\nabla}^2 \nabla^2 & 0 \\ 0 & \nabla^2 \bar{\nabla}^2 \end{pmatrix} \quad , \quad (6.5.47)$$

in terms of the square of  $\mathbf{O}$ . This action is given by  $S'_0 + \bar{S}'_0$ , where

$$S'_0 = \int d^4x d^2\theta \frac{1}{2} \Phi \square_+ \Phi = \int d^4x d^4\theta \frac{1}{2} \Phi \nabla^2 \Phi \quad . \quad (6.5.48)$$

In terms of it we can write the functional integral

$$\Delta^2 = \int \mathcal{D}\Phi \mathcal{D}\bar{\Phi} e^{S'_0 + \bar{S}'_0} = \left| \int \mathcal{D}\Phi e^{S'_0} \right|^2 = \left( \int \mathcal{D}\Phi e^{S'_0} \right)^2 . \quad (6.5.49)$$

We have used the fact that  $S'_0$  and its hermitian conjugate contribute equally to  $\Delta$ , as can be seen, for example, by examining the resulting Feynman rules below. (This procedure is analogous to the “doubling” trick in QED, where the analogue of  $\mathbf{O}$  is  $\nabla_{\alpha\beta}$ ; and of  $\square_+$  is  $C_{\alpha\beta}\square + f_{\alpha\beta}$ , with  $f_{\alpha\beta}$  the electromagnetic field strength. )

We now integrate  $S'_0$  by separating out  $\bar{D}^2 D^2$  from  $\bar{\nabla}^2 \nabla^2$  and treating  $(\bar{\nabla}^2 \nabla^2 - \bar{D}^2 D^2)$  as an interaction term. The result is

$$\Delta = e^{\int d^4x d^2\theta \frac{1}{2} \frac{\delta}{\delta j} [\bar{\nabla}^2 \nabla^2 - \bar{D}^2 D^2] \frac{\delta}{\delta j}} e^{-\int d^4x d^2\theta \frac{1}{2} j \square_0^{-1} j} . \quad (6.5.50)$$

(Writing instead  $\bar{\nabla}^2 \nabla^2 = \bar{D}^2 e^{-V} D^2 e^V$  in the chiral representation gives the rules for the one-loop expression in the usual noncovariant formalism.) Therefore, a calculation of the one-loop contribution consists in evaluating graphs with propagators  $p^{-2} \delta^4(\theta - \theta')$  and vertices  $\bar{\nabla}^2 \nabla^2 - \bar{D}^2 D^2$  giving rise to a string

$$\cdots [\bar{\nabla}^2 \nabla^2 - \bar{D}^2 D^2]_i \delta^4(\theta_i - \theta_{i+1}) [\bar{\nabla}^2 \nabla^2 - \bar{D}^2 D^2]_{i+1} \cdots , \quad (6.5.51)$$

with  $\int d^4\theta_i$  integrals at each vertex and one loop-momentum integral. We carry out the evaluation *in the chiral representation*, so that  $\bar{\nabla}_{\dot{\alpha}} = \bar{D}_{\dot{\alpha}}$ . We concentrate on the  $i$  vertex, and from the next vertex we temporarily transfer the  $\bar{D}^2 = \bar{\nabla}^2$  factor across the  $\delta$ -function. We now use the identity  $(\bar{\nabla}^2 \nabla^2 - \bar{D}^2 D^2) \bar{D}^2 = (\square_+ - \square_0) \bar{D}^2$  (in the chiral representation). Having performed this maneuver we return the  $\bar{D}^2$  to its original place, and proceed to manipulate the next vertex in the same way. This procedure can be carried out at all vertices but one, which retains its original form. The resulting rules for the evaluation of  $\Delta$  are, with the usual propagator,

one vertex:

$$\bar{D}^2 (\nabla^2 - D^2) , \quad (6.5.52)$$

other vertices:

$$\square_+ - \square_0 , \quad (6.5.53)$$

with the covariant derivatives in chiral representation. Now only one vertex contributes any  $\bar{D}$ 's and as a consequence the evaluation of one-loop graph contributions from chiral

superfields is considerably simplified. Higher loops are given by the rest of the expression in (6.5.44).

Up to now we have not specified whether the  $\nabla$ 's are full or background covariant derivatives. If both quantum and background gauge fields are present, it is more convenient to carry out the above procedure at an early stage, before we write  $\tilde{\Phi} = \bar{\Phi}e^V$ , i.e., work with fully covariantly chiral superfields (but not for the ghosts, which are only background covariantly chiral). The result of the calculation is expressible in terms of the full covariant derivatives, and only at that stage, having integrated out the chiral superfields, do we need to make the background quantum splitting on the gauge fields.

The “doubling” trick cannot be applied covariantly when the scalar multiplet is in a complex representation of the Yang-Mills group. If we write the covariantly chiral  $\Phi$  in terms of ordinary chiral  $\Phi_0$  ( $\bar{D}_\alpha \Phi_0 = 0$ ) in the vector representation

$$\Phi = e^{\bar{\Omega}} \Phi_0 \quad , \quad (6.5.54)$$

we have

$$\bar{\Phi} = e^{\bar{\Omega}^*} \bar{\Phi}_0 \quad , \quad (6.5.55)$$

and

$$\bar{\nabla}^2 \bar{\Phi} = e^{-\Omega^*} \bar{D}^2 e^{\Omega^*} e^{\bar{\Omega}^*} \bar{\Phi}_0 \quad , \quad (6.5.56)$$

is not in the same representation as  $\Phi$  (does not satisfy the same chirality condition) except when the representation is real (in which case  $\Omega^* = -\bar{\Omega}$ ). Therefore, the operator  $\mathbf{O}$  in (6.5.46) cannot be squared, since it is not representation-preserving. As a result, we must use rules *at one loop* which are not expressed manifestly in terms of connections  $\Gamma_A$ , but involve explicit gauge fields.

In (6.5.45) we express  $\Phi$  in terms of  $\Phi_0$ , and introduce ordinary chiral sources  $j_0$  ( $\bar{D}_\alpha j_0 = 0$ ). We have instead of (6.5.46) the following equations of motion in the presence of external super-Yang-Mills:

$$\hat{\mathbf{O}} \begin{pmatrix} \Phi_0 \\ \bar{\Phi}_0 \end{pmatrix} + \begin{pmatrix} j_0 \\ \bar{j}_0 \end{pmatrix} = 0 \quad , \quad \hat{\mathbf{O}} = \begin{pmatrix} 0 & \bar{D}^2 e^{V^*} \\ D^2 e^V & 0 \end{pmatrix} . \quad (6.5.57)$$

The noncovariant object  $\hat{\mathbf{O}}$  can be squared, since it preserves  $(\bar{D}_\alpha \cdot)$ -chirality:

$$\hat{\mathbf{O}}^2 = \begin{pmatrix} \bar{D}^2 e^{V*} D^2 e^V & 0 \\ 0 & D^2 e^V \bar{D}^2 e^{V*} \end{pmatrix} . \quad (6.5.58)$$

The action  $\hat{S}_0'$  obtained from  $\hat{\mathbf{O}}^2$  again gives a contribution equal to that of its hermitian conjugate. As in (6.5.50) we separate a  $D^2$  from  $e^{V*} D^2 e^V$  and treat the rest as an interaction. The propagator is as before, but the vertex is now

$$\bar{D}^2(e^{V*} D^2 e^V - D^2) . \quad (6.5.59)$$

Note that, for *real* representations,  $V^* = -\bar{V} = -V$ , so this vertex is just  $\bar{D}^2(\nabla^2 - D^2)$ , and the rules of (6.5.52,53) can be obtained. In general, for a group containing factors for which  $\Phi$  is in a real representation, we can write  $V = V_1 + V_2$ , where  $V_1^* = -V_1$ , but  $V_2^* \neq -V_2$  ( $[V_1, V_2] = 0$ ), and write the vertex as

$$\bar{D}^2(e^{V_2*} \nabla_1^2 e^{V_2} - D^2) , \quad \nabla_{1\alpha} = e^{-V_1} D_\alpha e^{V_1} = D_\alpha + \Gamma_{1\alpha} . \quad (6.5.60)$$

Then  $V_1$  appears in the rules only as  $\Gamma_{1A}$  while  $V_2$  appears explicitly.

The net result is that the effective action is expressed manifestly in terms of  $\nabla_A$  for Yang-Mills factors that occur coupled only to real representations, and *always* for higher-loop contributions. However, at one loop, and only for Yang-Mills factors coupled to complex representations, the contribution must be calculated in a way where the covariance is not manifest.

Our methods can also be applied to massive chiral superfields. In that case the term  $\bar{j} \square_+^{-1} j$  of (6.5.44) is replaced with

$$\bar{j} \frac{1}{\square_+ - m^2} j + \frac{1}{2} [j \frac{m \nabla^2}{\square_+ (\square_+ - m^2)} j + h.c.] , \quad (6.5.61)$$

a direct covariantization of the result (6.3.13). In performing the doubling trick, we use  $\Delta(m) = \Delta(-m)$ . We replace the kinetic operator

$$\mathbf{O}(m) = \begin{pmatrix} m & \bar{\nabla}^2 \\ \nabla^2 & m \end{pmatrix} \quad (6.5.62)$$

by